Forecasting Implied Volatility with HAR-RV Models

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Abstract

I develop a forecasting framework for one-day and five-day changes in implied volatility — proxied by the VIX — of the S&P 500 ETF (SPY), building on the Heterogeneous Autoregressive Realized Volatility (HAR-RV) model of Corsi (2009). The baseline specification is augmented with the variance risk premium (Carr & Wu, 2009), the VIX term-structure slope, and cross-asset covariates. Realized volatility is estimated from daily OHLC prices via the Garman–Klass estimator, which yields a more statistically efficient proxy than close-to-close returns without requiring proprietary tick data.

Six competing specifications are evaluated: a random walk, GARCH(1,1), HAR-OLS, Semi-HAR, HAR-Extended (Ridge), and HAR-Extended with regime interactions. All models are assessed through expanding-window walk-forward validation under strict real-time information constraints over the 2012–2024 sample. Pairwise predictive accuracy is tested with the Diebold–Mariano statistic and HLN finite-sample correction for non-nested comparisons, and with the Clark–West statistic for nested pairs; Benjamini–Hochberg FDR control is applied across the full test battery. Forecast calibration is examined via Mincer–Zarnowitz regressions.

On the theoretical side, I derive a prediction linking mean-reversion in the variance risk premium to a negative loading on ΔIV at the weekly horizon, providing the economic anchor for the five-day specification.

Keywords: implied volatility, HAR-RV, variance risk premium, Garman–Klass volatility, out-of-sample forecasting, Diebold–Mariano test, VIX term structure, walk-forward validation.

JEL Codes: C22, C53, G12, G13.

Contents

Nomenclature **5**

I Framework and Methodology **8**

1 Introduction **8**

2 Literature Review **9**
2.1 HAR-RV and Extensions . . . 9
2.2 Variance Risk Premium and Predictability . . . 10
2.3 Realized Volatility from OHLC Data . . . 10
2.4 Forecast Evaluation . . . 10

3 Data **11**
3.1 Sample Period . . . 11
3.2 Data Sources . . . 11
3.3 MOVE Index Proxy . . . 12
3.4 Credit Spread Construction . . . 13
3.5 Data Integrity and Reproducibility . . . 13
3.6 Data Cleaning . . . 13
3.7 Stationarity Pre-Testing . . . 14

4 Feature Engineering **14**
4.1 Prediction Targets . . . 15
4.2 Realized Volatility Estimators . . . 15
4.3 HAR Components . . . 16
4.4 Signed Semi-Variance Components . . . 17
4.5 Variance Risk Premium . . . 18
4.6 VIX Term Structure Features . . . 19
4.7 Macro and Cross-Asset Covariates . . . 20
4.8 Regime Classification . . . 20
4.9 Transformation Pipeline . . . 21

5 Model Specifications **22**
5.1 Model 0: Random Walk Benchmark . . . 22
5.2 Model 1: GARCH(1,1) Benchmark . . . 22
5.3 Model 2: HAR-OLS (Academic Baseline) . . . 24

5.4 Model 3: Semi-HAR (SHAR) 24
5.5 Model 4: HAR-Extended (Primary Specification) 25
5.6 Model 5: HAR with Selective Regime Interactions 25
5.7 Model Nesting Structure 27

6 Validation Design **27**
6.1 Expanding-Window Walk-Forward Protocol 27
6.2 Inner Cross-Validation for Ridge λ 28
6.3 Loss Functions and Evaluation Metrics 29
6.4 Out-of-Sample R^2 30
6.5 Directional Accuracy 30
6.6 Mincer–Zarnowitz Calibration 31
6.7 Forecast Comparison Tests 31
6.7.1 Diebold–Mariano Test with HLN Correction (Non-Nested) . . 31
6.7.2 Clark–West Test (Nested Models) 31
6.8 Multiple Testing Correction 32
6.9 Regime-Conditional Performance 33

II Empirical Results **34**

7 Exploratory Data Analysis **34**
7.1 Descriptive Statistics 34
7.2 Regime-Conditional Descriptive Statistics 35
7.3 Stationarity Tests 36
7.4 Distributional Features of the Targets 36
7.5 Time Series of Key Variables 37

8 In-Sample Estimation Results **39**
8.1 HAR-OLS Coefficient Estimates (Model 2) 39
8.2 SHAR Asymmetry Estimates (Model 3) 41
8.3 HAR-Extended Estimates (Model 4) 42
8.4 Residual Diagnostics and Structural Stability 43

9 Out-of-Sample Forecasting Performance **45**
9.1 Walk-Forward Performance Summary 45
9.2 Statistical Significance: Full Test Battery 47
9.3 Mincer–Zarnowitz Calibration 48
9.4 Temporal Dynamics of Forecast Performance 49

10 Regime-Conditional Analysis **50**

11 Robustness Analysis **51**
11.1 Sub-Period Stability 51
11.2 Alternative RV Estimators 52
11.3 Alternative Target Specifications 53
11.4 Training Window Configuration 53
11.5 Regularization Comparison 54
11.6 Regime Threshold Sensitivity 55
11.7 VRP Construction Sensitivity 55
11.8 Summary of Robustness 55

A Reproducibility **59**

Nomenclature

The following notation is used consistently throughout the paper. Scalars are set in italic, vectors in bold italic, and matrices in bold upright. All volatility quantities are annualized (multiply variance by 252) unless stated otherwise.

Price Data & Returns

O_t, H_t, L_t, C_t	Adjusted open, high, low, and close prices of SPY on day t.
r_t	Log return: $r_t = \log(C_t/C_{t-1})$.
$r_{t-1:t-5}$	Cumulative five-day lagged return: $\sum_{j=1}^{5} r_{t-j}$.

Realized Volatility

$\hat{\sigma}^2_{GK,t}$	Garman–Klass daily variance estimator (Eq. 5).
$\hat{\sigma}^2_{CC,t}$	Close-to-close daily variance: r_t^2.
$\hat{\sigma}^2_{P,t}$	Parkinson range-based daily variance (Eq. 8).
$\mathrm{RV}_t^{(h)}$	Annualized h-day realized volatility (Eq. 6).
$\mathrm{RV}_t^{(d)}$, $\mathrm{RV}_t^{(w)}$, $\mathrm{RV}_t^{(m)}$	HAR components: daily ($h = 1$), weekly ($h = 5$), monthly ($h = 22$).
RS_t^+, RS_t^-	Signed semi-variances (Eqs. 12–13).
$\mathrm{RV}_t^{(d,+)}$, $\mathrm{RV}_t^{(d,-)}$	Annualized upside/downside daily semi-volatilities (Eqs. 14–15).

Implied Volatility & Term Structure

VIX_t	CBOE 30-day implied volatility index (annualized, %).
$\mathrm{VIX9D}_t$	CBOE 9-day implied volatility index.
$\mathrm{VIX3M}_t$, $\mathrm{VIX6M}_t$	CBOE 3-month and 6-month implied volatility indices.
slope_t	Term-structure slope: $\mathrm{VIX3M}_t - \mathrm{VIX9D}_t$ (Eq. 19).
curv_t	Term-structure curvature: $\mathrm{VIX9D}_t - 2\,\mathrm{VIX}_t + \mathrm{VIX3M}_t$ (Eq. 20).

Risk Premia & Macro Variables

VRP_t	Variance Risk Premium: $\mathrm{VIX}_t^2 - [\mathrm{RV}_t^{(m)}]^2$ (Eq. 16).
Δy_t^{10Y}	Daily change in the 10-year Treasury yield (DGS10).

$\widehat{\text{MOVE}}_t$	Treasury volatility proxy: 22-day rolling annualized standard deviation of ΔDGS10 (Eq. 1).
spread_t	Credit spread: BBB minus AAA corporate bond yield (Eq. 2).

Targets & Forecast Objects

$y_t^{(h)}$	Prediction target: $\text{VIX}_{t+h} - \text{VIX}_t,\ \ h \in \{1, 5\}$.
$\hat{y}_t^{(h)}$	Model forecast of $y_t^{(h)}$.
h	Forecast horizon in trading days.

Probability & Measures

$\mathbb{P}$	Physical (statistical) probability measure.
$\mathbb{Q}$	Risk-neutral probability measure.
$\mathbb{E}_t[\,\cdot\,]$	Conditional expectation given filtration $\mathcal{F}_t$.
$\mathcal{F}_t$	Filtration: all market information available at or before close of day t.

GARCH Parameters

μ	Conditional mean of returns.
ω	GARCH intercept ($\omega > 0$).
α	ARCH coefficient (shock persistence).
β	GARCH coefficient (variance persistence).
σ_t^2	GARCH conditional variance at time t.
$\bar{\sigma}^2$	Unconditional variance: $\omega/(1-\alpha-\beta)$.
$s_t^{(h)}$	GARCH change-based IV signal (Eq. 27).

Regime & Regularization

$\mathcal{K}$	Regime set: $\{\text{Low},\ \text{Medium},\ \text{High}\}$.
$\mathbf{1}(\cdot)$	Indicator function.
λ	Ridge regularization parameter (ℓ_2 penalty).
$\boldsymbol{\theta}$	Parameter vector in regularized regression.
q^*	Andrews (1991) optimal HAC bandwidth (Eq. 30).

Evaluation Metrics

R^2_{OOS}	Out-of-sample R^2 relative to the random walk (M0).
MSE	Mean squared error.
QLIKE	Quasi-likelihood loss (Patton, 2011).

Abbreviations

CBOE	Chicago Board Options Exchange.
DM	Diebold–Mariano test.
CW	Clark–West test.
FDR	False Discovery Rate (Benjamini–Hochberg).
GK	Garman–Klass.
HAC	Heteroskedasticity and Autocorrelation Consistent.
HAR	Heterogeneous Autoregressive.
HLN	Harvey–Leybourne–Newbold (small-sample DM correction).
IV	Implied Volatility.
MZ	Mincer–Zarnowitz.
OLS	Ordinary Least Squares.
OHLC	Open, High, Low, Close.
QMLE	Quasi-Maximum Likelihood Estimation.
RV	Realized Volatility.
RW	Random Walk.
SHAR	Semi-HAR model.
VRP	Variance Risk Premium.

Part I

Framework and Methodology

1. Introduction

Implied volatility occupies a central role in asset pricing and risk management. Changes in implied volatility directly affect options pricing, delta-hedging costs, and portfolio risk budgets. Despite extensive research on equity return predictability, the short-horizon dynamics of implied volatility changes—what drives their direction and magnitude over one and five trading days—remain comparatively understudied.

This paper addresses a focused empirical question: *can observable financial variables, available at the close of each trading day, predict the subsequent change in implied volatility of SPY?* The question carries direct implications for options traders managing vega exposure, volatility arbitrageurs seeking to exploit mean-reversion in the variance risk premium, and risk managers calibrating short-term hedging strategies.

Theoretical mechanism. The Variance Risk Premium (VRP)—the wedge between risk-neutral and physical expectations of integrated variance—provides the economic foundation for weekly-horizon predictability. Investors consistently pay a premium for variance protection, making VRP positive on average. When VRP is elevated, options market-makers face increased inventory risk and respond by supplying volatility more aggressively, thereby exerting downward pressure on implied volatility. Proposition 4.1 formalizes this channel: under mean-reverting VRP dynamics, the conditional expectation of ΔVIX loads negatively on the current VRP level. This mechanism operates primarily at weekly frequencies—where options roll activity and delta-hedging rebalancing are concentrated—rather than at daily frequencies where microstructure noise dominates.

Methodological contributions. This study makes four methodological contributions:

1. **HAR-RV for implied volatility changes.** We repurpose the HAR-RV framework of Corsi (2009)—originally designed to forecast *realized* volatility—to predict *changes in implied* volatility, exploiting the information bridge between physical and risk-neutral measures.
2. **OHLC-based realized volatility.** Realized volatility is constructed from daily OHLC data via the Garman–Klass estimator (Garman & Klass, 1980), with explicit discussion of its bias properties under overnight gaps and jumps.
3. **Signed semi-variance decomposition.** We incorporate signed semi-variance components following Patton & Sheppard (2015), adapted to daily frequency with transparent acknowledgment of the proxy's limitations relative to the original high-frequency formulation.
4. **Strict out-of-sample discipline.** All models are evaluated under a rigorous out-of-sample protocol with real-time information constraints, purge gaps for overlapping targets, and nested hyperparameter selection—ensuring that no future information contaminates any forecast.

2. Literature Review

This section situates the paper within four intersecting strands of the literature: the HAR-RV modeling family, variance risk premium predictability, OHLC-based realized volatility estimation, and out-of-sample forecast evaluation methodology.

2.1 HAR-RV and Extensions

The Heterogeneous Autoregressive model of Corsi (2009), motivated by the Heterogeneous Market Hypothesis of Müller et al. (1993), decomposes realized volatility into daily, weekly, and monthly components—each capturing the trading horizon of a distinct class of market participants. Despite its parsimonious OLS-estimable structure, the model successfully approximates the long-memory property that characterizes financial volatility (Andersen et al., 2003).

Patton & Sheppard (2015) extend HAR-RV with signed semi-variances, decomposing the daily component into contributions from positive and negative intraday returns. Their Semi-HAR (SHAR) model documents asymmetric persistence—a leverage effect operating within the volatility process itself—and dominates HAR-OLS out-of-sample for major equity indices. Critically, their decomposition relies on 5-minute returns; our daily-frequency adaptation constitutes a substantially coarser proxy, with implications discussed in Section 4.4.

Corsi & Renò (2012) augment HAR-RV with a leverage component (lagged returns × realized variance). For SPY, where the equity-volatility correlation is approximately −0.7, this extension carries economically meaningful content.

2.2 Variance Risk Premium and Predictability

Carr & Wu (2009) establish the VRP as the fair price of a variance swap and document its pronounced time-variation and mean-reversion. Bollerslev et al. (2009) demonstrate that VRP predicts aggregate equity returns at quarterly horizons within the Long-Run Risk framework of Bansal Yaron. Bekaert & Hoerova (2014) decompose VIX^2 into a conditional variance component and a risk-aversion component, finding that VRP retains predictive power for returns even after controlling for realized variance.

Notably, neither Carr & Wu (2009) nor Bollerslev et al. (2009) targets predictability of *implied volatility changes* at short horizons. Christensen & Prabhala (1998) study implied volatility as a forecast of future realized volatility, but not the reverse direction—i.e., using physical-measure information to predict risk-neutral-measure dynamics. This gap is precisely what the present paper addresses.

2.3 Realized Volatility from OHLC Data

Parkinson (1980) shows that the high-low range estimator achieves approximately 5× the efficiency of close-to-close squared returns under geometric Brownian motion. Garman & Klass (1980) improve this to ∼7.4× efficiency by incorporating all four OHLC prices. However, both estimators are *downward biased* in the presence of overnight price gaps—since the open-to-previous-close return falls outside the observed range—and *upward biased* by discrete price jumps that inflate the high-low spread. Yang & Zhang (2000) propose a drift-independent estimator that combines overnight and intraday components, partially addressing both limitations; we include it as a robustness check in Section 11.

2.4 Forecast Evaluation

Hansen (2005) documents the well-known difficulty of outperforming GARCH(1,1) in out-of-sample forecasting exercises, motivating its inclusion in our model hierarchy as a non-trivial benchmark. Patton (2011) demonstrates that MSE is not robust to noise in the volatility proxy, whereas QLIKE preserves correct model rankings; we report both loss functions. Campbell & Thompson (2008) formalize the out-of-sample R^2 statistic (R^2_{OOS}) that serves as our primary evaluation metric. Clark & West (2006) derive the appropriate test statistic for nested out-of-sample forecast comparisons, correcting the well-known downward size distortion of standard

Diebold–Mariano tests in nested settings. Finally, Giacomini & White (2006) provide the conditional predictive ability framework that underpins our expanding-window evaluation protocol.

3. Data

3.1 Sample Period

The dataset spans January 4, 2010 to December 31, 2024, comprising $T \approx 3{,}773$ trading days. The sample start date is chosen to exclude the 2008–2009 Global Financial Crisis, which constitutes a structural outlier that would distort coefficient estimates and impose unrealistic training conditions on the walk-forward protocol. The first approximately 500 trading days (2010–2011) serve as the initial training window; out-of-sample evaluation commences in early 2012.

3.2 Data Sources

All data are obtained from two publicly accessible sources: Yahoo Finance (equity and volatility index prices) and the Federal Reserve Economic Data (FRED) system (interest rates and credit spreads). No proprietary or subscription-based data are required, ensuring full reproducibility. Table 11 itemizes each data feed, including tickers, source identifiers, and coverage periods.

VIX9D availability. The CBOE 9-day VIX (`^VIX9D`) is available from January 2011, approximately 252 trading days after the sample start. For the pre-2011 period, VIX9D is proxied by VIX itself. Since both indices measure near-term implied volatility (9-day vs. 30-day), the term-structure slope $\text{slope}_t = \text{VIX3M}_t - \text{VIX9D}_t$ remains a valid—if attenuated—measure of the 3-month versus short-term IV spread, with the proxy reducing slope sensitivity at the short end. This imputation ensures sample alignment across all models ($T_{\text{oos}} \approx 3{,}250$ for M2–M5) and avoids the loss of 252 observations that would otherwise affect models using term-structure features.

Table 11: Data Sources, Access Methods, and Sample Coverage

Variable	Ticker / Series ID	Source	Freq.	Period
Panel A: Equity & Volatility		*Yahoo Finance (`auto_adjust=True`)*		
SPY OHLCV (adjusted)	`SPY`	Yahoo	D	2010–2024
VIX (30-day IV)	`^VIX`	CBOE	D	2010–2024
VIX9D (9-day IV)	`^VIX9D`	CBOE	D	2011–2024[a]
VIX3M (3-month IV)	`^VIX3M`	CBOE	D	2010–2024
VIX6M (6-month IV)	`^VIX6M`	CBOE	D	2010–2024
Panel B: Macro & Fixed Income		*FRED (public CSV API)*		
10-Year Treasury yield	`DGS10`	FRED	D	2010–2024
AAA corporate bond yield	`BAMLC0A1CAAA`	FRED	D	2010–2024
BBB corporate bond yield	`BAMLC0A4CBBB`	FRED	D	2010–2024
Panel C: Derived Variables		*Computed from Panels A–B*		
Credit spread	BBB − AAA	Eq. (2)	D	2010–2024
Treasury vol. (MOVE proxy)	22-day vol of ΔDGS10	Eq. (1)	D	2010–2024

[a] VIX9D available from January 2011; earlier observations proxied by VIX (see text).

Notes: D = Daily (NYSE trading days). All CBOE indices accessed via Yahoo Finance. Yahoo `auto_adjust=True` applies a uniform split / dividend ratio to O, H, L, C, preserving intraday ranges for Garman–Klass estimation. FRED missing values on bank holidays are forward-filled (last observation carried forward); no interpolation is applied. Each download is logged with a SHA-256 checksum for reproducibility verification.

3.3 MOVE Index Proxy

The ICE BofA MOVE Index—the standard measure of Treasury implied volatility—is not available through free public APIs. We construct a realized-volatility-based proxy using the 22-day rolling standard deviation of daily changes in the 10-year Treasury yield, annualized:

$$\widehat{\mathrm{MOVE}}_t = \sqrt{252} \times \mathrm{Std}\Big(\Delta\mathrm{DGS10}_{t-21},\ \ldots,\ \Delta\mathrm{DGS10}_t\Big), \tag{1}$$

where $\Delta\mathrm{DGS10}_s \equiv \mathrm{DGS10}_s - \mathrm{DGS10}_{s-1}$. This proxy captures the same economic content as MOVE—volatility of interest rates at the 10-year tenor—but substitutes realized for implied volatility. The correlation between this proxy and the actual MOVE index (when both are available) typically exceeds $\rho > 0.85$ at daily frequency.

3.4 Credit Spread Construction

The credit spread is computed as the difference between BBB and AAA corporate bond yields from the ICE BofA indices:

$$\text{spread}_t = \text{BBB}_t - \text{AAA}_t \,, \tag{2}$$

where BBB_t denotes the effective yield on the ICE BofA BBB US Corporate Index (`BAMLC0A4CBBB`) and AAA_t is the corresponding AAA index (`BAMLC0A1CAAA`). This spread captures aggregate credit risk appetite and widens markedly during periods of financial stress, providing information complementary to equity-based volatility measures.

3.5 Data Integrity and Reproducibility

Each download is logged in a JSON manifest file containing: (i) UTC timestamp, (ii) number of observations, and (iii) SHA-256 checksum of the cached `parquet` file. A verification function compares stored checksums against current files to detect retroactive adjustments by data providers—particularly Yahoo Finance's split and dividend corrections, which can silently alter historical values. If any checksum mismatch is detected, the affected series is re-downloaded and re-cached before proceeding.

3.6 Data Cleaning

1. **Calendar alignment.** All series are aligned to the SPY trading calendar (NYSE business days). FRED series, which may exhibit gaps on bank holidays, are forward-filled to match. No backward-fill or interpolation is applied.
2. **SPY price adjustments.** Yahoo Finance's `auto_adjust` flag applies split and dividend adjustments uniformly to all four OHLC prices using a single multiplicative ratio, thereby preserving the intraday high-low range required for unbiased Garman–Klass estimation.
3. **Outlier treatment.** No observations are removed. COVID-19 (March 2020), the 2022 rate-hiking shock, and the August 2024 VIX spike are retained as part of the data-generating process. Sensitivity to these episodes is assessed in Section 11.
4. **Missing data.** After forward-fill, any remaining missing values (arising from series start dates or residual data gaps) produce `NaN` rows that are dropped during feature-matrix construction. The effective sample size for each model depends on its feature set and is reported in the results tables.

3.7 Stationarity Pre-Testing

VIX in levels exhibits high persistence but mean-reverts over multi-week horizons. Our prediction target $y_t^{(h)} = \Delta\text{VIX}_{t,h}$ is the first difference, which standard unit root tests—both Augmented Dickey–Fuller (ADF) and Kwiatkowski–Phillips–Schmidt–Shin (KPSS)—confirm is $I(0)$ at any conventional significance level. All regressors—HAR components, VRP, slope, curvature, spreads, returns—are either stationary by construction (returns, spreads, first differences) or bounded and mean-reverting. VIX levels enter only through VRP and term-structure combinations, which are differences of co-integrated series and hence stationary. Formal test results are reported in Table 17 (Section 7).

Remark 3.1 (Stambaugh bias)**.** Persistent regressors (VRP, credit spread, MOVE proxy) in predictive regressions induce the Stambaugh (1999) finite-sample bias, which tilts inference toward rejection of the no-predictability null. This concern applies to in-sample inference but does *not* affect out-of-sample forecast evaluation—our primary assessment tool. In-sample t-statistics should accordingly be interpreted with this caveat in mind.

Remark 3.2 (VIX as implied volatility proxy)**.** VIX is formally defined on S&P 500 index options, not SPY ETF options. In practice, the two track each other with a correlation exceeding $\rho > 0.99$ in levels; we treat the approximation as immaterial for daily forecasting purposes.

4. Feature Engineering

All features are computed by a deterministic, stateless transformation pipeline applied to the raw data described in Section 3. The single inviolable constraint governing the entire pipeline is:

> *Every feature at index t is a measurable function of the filtration $\mathcal{F}_t$—i.e., market data observable at or before the close of day t. Features are paired with the target $y_t^{(h)} = \text{VIX}_{t+h} - \text{VIX}_t$; no information from days $t+1, \ldots, t+h$ enters any feature computation.*

4.1 Prediction Targets

We define two prediction targets as arithmetic changes in VIX:

$$y_t^{(1)} = \mathrm{VIX}_{t+1} - \mathrm{VIX}_t, \tag{3}$$

$$y_t^{(5)} = \mathrm{VIX}_{t+5} - \mathrm{VIX}_t. \tag{4}$$

Arithmetic differences are chosen over log differences because they preserve direct interpretability when the VRP is measured in variance units. Log and percentage difference targets are evaluated as robustness checks (Section 11).

In the implementation, targets are constructed via `vix.shift(-h) - vix`, which produces `NaN` for the last h rows. These rows are excluded during model fitting—they represent dates whose targets have not yet been realized.

Remark 4.1 (Overlapping observations for $h = 5$)**.** The five-day target induces MA(4) serial correlation in the forecast errors: $y_t^{(5)}$ and $y_{t+1}^{(5)}$ share four common trading days. Consequently, all standard errors and test statistics for $h = 5$ employ Newey–West HAC estimation with at least $h - 1 = 4$ lags. In practice, we use the Andrews (1991) data-dependent bandwidth q^* (Eq. 30), which exceeds this minimum throughout the sample.

4.2 Realized Volatility Estimators

Let O_t, H_t, L_t, C_t denote the adjusted open, high, low, and close prices of SPY on day t, and let $r_t \equiv \log(C_t/C_{t-1})$ denote the close-to-close log return.

Definition 4.1 (Garman–Klass Daily Variance—Primary Estimator)**.**

$$\hat{\sigma}^2_{GK,t} = \tfrac{1}{2}\Big(\log H_t - \log L_t\Big)^2 - (2\log 2 - 1)\Big(\log C_t - \log O_t\Big)^2. \tag{5}$$

Values are clipped at zero to handle the rare numerical case where $C_t \approx O_t$ and $H_t \approx L_t$ produce a small negative result due to finite precision.

Definition 4.2 (Annualized h-Day Realized Volatility)**.**

$$\mathrm{RV}_t^{(h)} = \sqrt{\frac{252}{h}\sum_{i=1}^{h} \hat{\sigma}^2_{GK,\,t-i+1}}. \tag{6}$$

The sum runs over the window $\{t - h + 1, \ldots, t\}$, ending at day t and using only information available at the close of day t. The factor $252/h$ annualizes the average daily variance.

For robustness, two alternative daily variance estimators are also computed:

$$\hat{\sigma}^2_{CC,t} = r_t^2 \qquad \Longrightarrow \qquad \mathrm{RV}_t^{(h,CC)} = \sqrt{\tfrac{252}{h} \sum_{i=1}^{h} r^2_{t-i+1}}, \tag{7}$$

$$\hat{\sigma}^2_{P,t} = \frac{(\log H_t - \log L_t)^2}{4 \log 2} \qquad \Longrightarrow \qquad \mathrm{RV}_t^{(h,P)} = \sqrt{\tfrac{252}{h} \sum_{i=1}^{h} \hat{\sigma}^2_{P,\,t-i+1}}. \tag{8}$$

Estimator properties and limitations. Under geometric Brownian motion with no overnight gaps, $\hat{\sigma}^2_{GK}$ is approximately 7.4× more efficient than r_t^2 (Garman & Klass, 1980). Three sources of bias arise when the estimator is applied to real equity data:

1. **Overnight gap omission.** GK uses only the open-to-close range. SPY exhibits non-trivial overnight variance (approximately 15–25% of total daily variance), which GK systematically excludes, producing *downward bias*.
2. **Jump sensitivity.** Large intraday jumps inflate the high-low range, creating *upward bias* during event-driven sessions (e.g., FOMC announcements, flash crashes).
3. **Drift bias.** GK assumes zero drift within the trading day. On strongly trending sessions, non-zero drift inflates measured variance relative to the true diffusive component.

These biases partially offset one another. For daily-frequency OHLC data on a liquid ETF, GK remains the preferred estimator in the absence of tick-level data.

Implementation note. The annualized RV is computed as

```
sqrt(252 / h * daily_var.rolling(h).sum())
```

using `.rolling().sum()` rather than `.rolling().std()`. The latter would return `NaN` for $h = 1$ due to the default `ddof=1` denominator in pandas.

4.3 HAR Components

The three components of the Heterogeneous Autoregressive model (Corsi, 2009) decompose realized volatility by the characteristic horizon of distinct market partici-

pants:

$$\mathrm{RV}_t^{(d)} = \mathrm{RV}_t^{(1)}, \qquad \text{[daily: high-frequency traders]} \tag{9}$$

$$\mathrm{RV}_t^{(w)} = \frac{1}{5}\sum_{j=0}^{4} \mathrm{RV}_{t-j}^{(1)}, \qquad \text{[weekly: institutional rebalancers]} \tag{10}$$

$$\mathrm{RV}_t^{(m)} = \frac{1}{22}\sum_{j=0}^{21} \mathrm{RV}_{t-j}^{(1)}. \qquad \text{[monthly: portfolio managers]} \tag{11}$$

In the implementation, $\mathrm{RV}^{(w)}$ and $\mathrm{RV}^{(m)}$ are computed as rolling means of $\mathrm{RV}^{(d)}$: `rv_d.rolling(5).mean()` and `rv_d.rolling(22).mean()`. This produces `NaN` for the first 21 rows, which are dropped during feature-matrix assembly.

Remark 4.2 (Multicollinearity)**.** By construction, $\mathrm{RV}^{(w)}$ contains $\mathrm{RV}^{(d)}$ as one of its five summands, and $\mathrm{RV}^{(m)}$ contains both shorter components. The resulting multicollinearity inflates OLS standard errors but does not bias coefficient estimates. Under Ridge regularization (Models 4 and 5), the ℓ_2 penalty shrinks correlated coefficients jointly rather than arbitrarily zeroing one of a collinear pair—as Lasso would—making Ridge the natural choice for this feature structure (Hoerl & Kennard, 1970).

4.4 Signed Semi-Variance Components

Following Patton & Sheppard (2015), we decompose daily squared returns into upside and downside contributions based on the sign of the close-to-close return:

$$\mathrm{RS}_t^{+} = r_t^2 \cdot \mathbf{1}(r_t \geq 0), \tag{12}$$

$$\mathrm{RS}_t^{-} = r_t^2 \cdot \mathbf{1}(r_t < 0), \tag{13}$$

so that the identity $r_t^2 = \mathrm{RS}_t^{+} + \mathrm{RS}_t^{-}$ holds exactly (verified via assertion in the code). The annualized daily semi-volatilities are:

$$\mathrm{RV}_t^{(d,+)} = \sqrt{252 \cdot \mathrm{RS}_t^{+}}, \tag{14}$$

$$\mathrm{RV}_t^{(d,-)} = \sqrt{252 \cdot \mathrm{RS}_t^{-}}. \tag{15}$$

On negative-return days, $\mathrm{RV}_t^{(d,+)} = 0$ and $\mathrm{RV}_t^{(d,-)} > 0$; on positive-return days, the reverse holds. Exactly one of the two components is nonzero on any given trading day.

Remark 4.3 (Proxy quality relative to the high-frequency original)**.** The original Patton & Sheppard (2015) decomposition accumulates signed 5-minute intraday

returns, producing a granular separation of upside and downside variation within each day. Our daily-frequency adaptation reduces each day to a *single* sign classification, discarding within-day asymmetry information entirely. For example, a session with large intraday swings in both directions but a small positive close-to-close return is classified as entirely "upside." The empirical content of the SHAR specification (Model 3) should be interpreted accordingly: weak out-of-sample results may reflect proxy coarseness rather than absence of the underlying leverage asymmetry.

4.5 Variance Risk Premium

Definition 4.3 (Variance Risk Premium)**.**

$$\mathrm{VRP}_t = \underbrace{\mathrm{VIX}_t^2}_{\approx\ \mathbb{E}_t^{\mathbb{Q}}\left[\int_t^{t+30/252}\sigma_s^2\,ds\right]\ \times\ 252} - \underbrace{\left[\mathrm{RV}_t^{(m)}\right]^2}_{\text{backward-looking proxy for } \mathbb{E}_t^{\mathbb{P}}[\,\cdot\,]}, \tag{16}$$

where both terms are expressed in **annualized variance points** (units: $\%^2$/year). Numerical example: when $\mathrm{VIX} = 20$ and $\mathrm{RV}^{(m)} = 15$, we obtain $\mathrm{VRP} = 400 - 225 = 175$.

$\mathrm{VRP}_t > 0$ on average reflects the premium that investors pay for variance protection (Carr & Wu, 2009). The backward-looking realized variance proxy introduces classical errors-in-variables attenuation, biasing estimated coefficients toward zero; any significance found is therefore conservative.

Proposition 4.1 (VRP Mean-Reversion Implies $\beta_{\mathrm{VRP}} < 0$)**.** Under three assumptions:

(i) $\mathrm{VIX}_t^2 = \mathbb{E}_t^{\mathbb{P}}[\mathrm{IV}_{t,t+\tau}] + \mathrm{VRP}_t$, where $\mathrm{IV}_{t,t+\tau}$ denotes integrated variance over the interval $[t,\, t+\tau]$;

(ii) VRP follows a stationary AR(1) process: $\mathrm{VRP}_{t+h} = \bar{\psi} + \rho\,(\mathrm{VRP}_t - \bar{\psi}) + \eta_{t+h}$, with $|\rho| < 1$;

(iii) $\mathbb{E}_t^{\mathbb{P}}[\mathrm{IV}_{t+h,\,t+h+\tau}]$ is approximately a martingale with respect to VRP dynamics;

then the conditional expectation of the squared-VIX change satisfies:

$$\mathbb{E}_t\left[\mathrm{VIX}_{t+h}^2 - \mathrm{VIX}_t^2\right] \approx (\rho^h - 1)\,(\mathrm{VRP}_t - \bar{\psi}), \tag{17}$$

with $(\rho^h - 1) < 0$ for $\rho < 1$. Linearizing via a first-order Taylor expansion for small changes:

$$\mathbb{E}_t[\Delta\mathrm{VIX}_{t,h}] \approx \frac{\rho^h - 1}{2\,\mathrm{VIX}_t}\,(\mathrm{VRP}_t - \bar{\psi}), \tag{18}$$

yielding $\beta_{\mathrm{VRP}} < 0$ in a regression of $\Delta\mathrm{IV}$ on VRP_t. The predictive effect strengthens

with the forecast horizon h and weakens as $\rho \to 1$ (i.e., as VRP persistence increases).

Proof sketch. By assumption (i),

$$\mathrm{VIX}^2_{t+h} = \mathbb{E}^{\mathbb{P}}_{t+h}\Big[\mathrm{IV}_{t+h,\,t+h+\tau}\Big] + \mathrm{VRP}_{t+h}\,.$$

Taking $\mathbb{E}_t[\,\cdot\,]$ and applying the martingale approximation (iii):

$$\mathbb{E}_t\Big[\mathrm{VIX}^2_{t+h}\Big] \approx \mathbb{E}^{\mathbb{P}}_t\Big[\mathrm{IV}_{t,\,t+\tau}\Big] + \mathbb{E}_t[\mathrm{VRP}_{t+h}]\,.$$

Substituting the AR(1) conditional expectation from (ii),

$$\mathbb{E}_t[\mathrm{VRP}_{t+h}] = \bar{\psi} + \rho^h\Big(\mathrm{VRP}_t - \bar{\psi}\Big),$$

and differencing with $\mathrm{VIX}^2_t = \mathbb{E}^{\mathbb{P}}_t[\mathrm{IV}_{t,\,t+\tau}] + \mathrm{VRP}_t$ yields (17).

The linearization in (18) follows from the first-order approximation

$$\sqrt{x+\delta} - \sqrt{x} \;\approx\; \frac{\delta}{2\sqrt{x}}$$

applied to VIX^2. □

Remark 4.4 (Role of Assumption (iii))**.** Assumption (iii) is the strong identifying restriction. If physical expected variance is itself predictable—as the HAR structure posits—then the HAR components in Models 2–5 absorb that variation, and β_{VRP} captures the *incremental* contribution of risk-premium mean-reversion, conditional on the physical variance forecast.

4.6 VIX Term Structure Features

Two features capture the shape of the VIX implied volatility term structure:

$$\mathrm{slope}_t = \mathrm{VIX3M}_t - \mathrm{VIX9D}_t, \tag{19}$$

$$\mathrm{curv}_t = \mathrm{VIX9D}_t - 2\cdot\mathrm{VIX}_t + \mathrm{VIX3M}_t. \tag{20}$$

- **Slope** (long minus short): $\mathrm{slope}_t > 0$ indicates contango (the normal state); $\mathrm{slope}_t < 0$ indicates backwardation (acute stress). Backwardation signals elevated near-term fear relative to medium-term expectations and is typically associated with rapid mean-reversion in implied volatility.
- **Curvature** (second difference): Captures a "humped" term structure in which the 30-day VIX exceeds the linear interpolation between the 9-day and 3-month points, signaling localized risk concentration at the one-month horizon.

Both features require VIX9D, which is available from approximately January 2011. Observations before that date produce `NaN` and are excluded from models that incorporate term-structure features (M4).

4.7 Macro and Cross-Asset Covariates

Five covariates provide information beyond the equity volatility complex:

- **Treasury yield change:** $\Delta y_t^{10Y} = \text{DGS10}_t - \text{DGS10}_{t-1}$. Sharp yield increases signal tightening financial conditions and historically co-move with equity volatility spikes.
- **Treasury volatility proxy:** $\widehat{\text{MOVE}}_t$ (Eq. 1), the 22-day rolling annualized standard deviation of ΔDGS10. Captures cross-asset volatility transmission (rates $\rightarrow$ equities), providing information orthogonal to equity-only features.
- **Credit spread:** $\text{spread}_t = \text{BBB}_t - \text{AAA}_t$ (Eq. 2), using ICE BofA corporate bond indices. Widens during financial stress; proxies aggregate credit risk appetite and funding conditions.
- **Lagged daily return:** $r_{t-1} = \log(C_{t-1}/C_{t-2})$. The explicit `shift(1)` ensures the feature at index t uses only information from day $t-1$ (filtration $\mathcal{F}_{t-1}$).
- **Lagged weekly return:** $r_{t-1:t-5} = \sum_{j=1}^{5} r_{t-j}$, computed as `ret.rolling(5).sum().shift(1)`. Together with the daily lag, this feature captures the well-documented asymmetric return–volatility relationship at short horizons.

All covariates satisfy the filtration constraint: Δy_t^{10Y}, $\widehat{\text{MOVE}}_t$, and spread_t are observable at the close of day t; both return features use information from day $t-1$ or earlier.

4.8 Regime Classification

A three-state volatility regime is defined using fixed VIX thresholds:

$$\text{regime}_t = \begin{cases} \text{Low} & \text{if } \text{VIX}_t < 15, \\ \text{Medium} & \text{if } 15 \leq \text{VIX}_t < 25, \\ \text{High} & \text{if } \text{VIX}_t \geq 25. \end{cases} \tag{21}$$

Implemented via:

```
pd.cut(vix, bins=[-inf, 15, 25, inf], labels=['Low', 'Medium',
                                              'High'])
```

The regime variable is categorical and enters Model 5 exclusively through indicator

interactions, not as a numerical feature..

Remark 4.5 (Threshold justification)**.** Thresholds are fixed *ex ante*, not estimated from data, to avoid look-ahead bias. The values 15 and 25 correspond approximately to the 30th and 80th percentiles of the VIX distribution over 1990–2009 (the pre-sample period) and align with conventional practitioner classifications. Sensitivity to alternative cutoffs (12 / 20 and 18 / 30) and to rolling data-driven terciles is assessed in Section 11.

4.9 Transformation Pipeline

The feature engineering pipeline executes the following ten steps sequentially:

1. **Daily variance.** Compute $\hat{\sigma}^2_{GK,t}$ from SPY OHLC (Eq. 5); clip negative values at zero.
2. **HAR triple.** Compute $\mathrm{RV}^{(d)}$, $\mathrm{RV}^{(w)}$, $\mathrm{RV}^{(m)}$ via rolling windows of 1, 5, 22 days (Eqs. 9–11).
3. **Semi-variances.** Compute $\mathrm{RV}^{(d,+)}$ and $\mathrm{RV}^{(d,-)}$ from close-to-close returns (Eqs. 14–15); assert $\mathrm{RS}^+ + \mathrm{RS}^- = r^2$.
4. **Variance risk premium.** Compute
$$\mathrm{VRP}_t = \mathrm{VIX}_t^2 - \left[\mathrm{RV}_t^{(m)}\right]^2 \qquad \text{(Eq. 16)}.$$
5. **Term structure.** Compute slope and curvature from VIX9D, VIX, VIX3M (Eqs. 19–20).
6. **Macro covariates.** Compute Δy^{10Y}, MOVE proxy (Eq. 1), and credit spread (Eq. 2).
7. **Lagged returns.** r_{t-1} via `shift(1)`; $r_{t-1:t-5}$ via `rolling(5).sum().shift(1)`.
8. **Regime.** Classify VIX level into Low / Medium / High (Eq. 21).
9. **Targets.** Compute $y^{(1)}$, $y^{(5)}$ via `shift(-h)` (Eqs. 3–4).
10. **Assembly.** Join all features and targets by date index; drop any row containing `NaN` (arising from rolling-window warm-up periods or series start-date mismatches).

The effective sample size after step 10 depends on the longest rolling window (22 days for $\mathrm{RV}^{(m)}$) and the latest series start date (VIX9D, approximately January 2011, for models using term-structure features).

Table 12: Feature Utilization by Model Specification

	HAR Core			**Semi-Var.**		**Vol Term Struct.**			**Macro**			**Returns**		**Regime**
Model	d	w	m	d^+	d^-	VRP	slp	crv	Δy	MOV	spr	r_1	r_{15}	R
M0 (Random Walk)														
M1 (GARCH)														
M2 (HAR)	✓	✓	✓											
M3 (SHAR)	✓	✓	✓	✓	✓									
M4 (Kitchen Sink)	✓	✓	✓			✓	✓	✓	✓	✓	✓	✓	✓	
M5 (Regime)	✓	✓	✓			✓								✓

Notes: HAR Core = $\{\mathrm{RV}^{(d)}, \mathrm{RV}^{(w)}, \mathrm{RV}^{(m)}\}$. Semi-Var. = realized semivariance components $\{\mathrm{RV}^{(d,+)}, \mathrm{RV}^{(d,-)}\}$. Vol Term Struct. = {VRP, slope, curvature}. Macro = $\{\Delta y^{10Y}, \widehat{\mathrm{MOVE}}, \text{spread}\}$. Returns = lagged daily and weekly returns. Empty cells indicate the feature is not used by that model.

5. Model Specifications

Six specifications (M0–M5) form a nested hierarchy of progressively richer information sets, summarized in Table 13. Let $y_t^{(h)} = \mathrm{VIX}_{t+h} - \mathrm{VIX}_t$ denote the h-step-ahead arithmetic VIX change and ε_t a mean-zero error term satisfying $\mathbb{E}_t[\varepsilon_t] = 0$. All models are evaluated at horizons $h \in \{1, 5\}$.

5.1 Model 0: Random Walk Benchmark

$$\hat{y}_t^{(h)} = 0, \qquad \forall\, t,\, h. \tag{22}$$

The random walk embodies the null hypothesis of no predictability: the best forecast of any future implied volatility change is identically zero. Every competing model must produce statistically significant improvements over M0 to be considered useful. The out-of-sample R^2_{OOS} relative to M0 serves as the primary evaluation metric throughout the paper.

5.2 Model 1: GARCH(1,1) Benchmark

The GARCH(1,1) model specifies the conditional mean and variance of SPY log returns as:

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t\, z_t, \qquad z_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1), \tag{23}$$

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2, \qquad \omega > 0,\ \ \alpha, \beta \geq 0,\ \ \alpha + \beta < 1. \tag{24}$$

Parameters $\boldsymbol{\psi} = (\mu,\, \omega,\, \alpha,\, \beta)^\top$ are estimated by Quasi-Maximum Likelihood (QMLE) in each expanding training window using the `arch` library.

Multi-step forecast. The j-step-ahead conditional variance follows the recursion:

$$\mathbb{E}_t\Big[\sigma^2_{t+j}\Big] = \bar{\sigma}^2 + (\alpha+\beta)^{j-1}\Big(\sigma^2_{t+1|t} - \bar{\sigma}^2\Big), \qquad j = 1,\, 2,\, \dots\,, \tag{25}$$

where $\bar{\sigma}^2 = \omega \,/\, (1-\alpha-\beta)$ denotes the unconditional variance. The average conditional variance over the forecast window $[t+1,\ t+h]$ admits the closed-form expression:

$$\bar{\sigma}^2_{t,h} = \bar{\sigma}^2 + \frac{\sigma^2_{t+1|t} - \bar{\sigma}^2}{h} \cdot \frac{1-(\alpha+\beta)^h}{1-(\alpha+\beta)}. \tag{26}$$

Change-based implied volatility signal. A naïve level mapping $\hat{y}_t = \sqrt{252\,\bar{\sigma}^2_{t,h}} - \mathrm{VIX}_t$ would equate $\mathbb{P}$-variance with $\mathbb{Q}$-implied variance, producing a systematic negative bias equal to the time-varying variance risk premium (see Remark 5.1). To remove this level mismatch while preserving the GARCH information about *changes* in conditional variance, we construct a raw signal based on the predicted change in GARCH-implied annualized volatility:

$$s_t^{(h)} = \sqrt{252 \cdot \bar{\sigma}^2_{t,h}} \;-\; \sqrt{252 \cdot \sigma^2_{t|t-1}}, \tag{27}$$

where $\sigma^2_{t|t-1}$ is the current one-step conditional variance (known at time t) and $\bar{\sigma}^2_{t,h}$ is the h-step average from (26). The signal $s_t^{(h)}$ measures how much the GARCH model expects annualized volatility to change over the next h days, expressed in the same units as VIX (percentage points of annualized volatility).

Mincer–Zarnowitz calibration. The raw signal is mapped to ΔVIX via a linear calibration estimated exclusively on training data:

$$\hat{y}_t^{(h)} = \hat{a}_l + \hat{b}_l\, s_t^{(h)}, \qquad (\hat{a}_l,\, \hat{b}_l) = \arg\min_{a,\,b} \sum_{\tau=1}^{T_l}\Big(y_\tau - a - b\, s_\tau^{(h)}\Big)^2, \tag{28}$$

where T_l denotes the training-set size at walk-forward fold l. This Mincer–Zarnowitz regression serves two distinct purposes:

(i) the intercept $\hat{a}_l$ absorbs any residual mean bias, including the average VRP effect;

(ii) the slope $\hat{b}_l$ rescales $\mathbb{P}$-measure volatility changes to match the typical magnitude of $\mathbb{Q}$-implied changes, since VIX—being a 30-day risk-neutral expectation—responds to shocks with a different elasticity than GARCH-filtered realized

volatility.

Both coefficients are re-estimated at each fold, preserving the strict temporal discipline of the walk-forward protocol.

Remark 5.1 (Measure mismatch and the role of calibration)**.** GARCH models σ_t^2 under the physical measure $\mathbb{P}$, whereas VIX reflects risk-neutral expectations under $\mathbb{Q}$. The variance risk premium $\mathrm{VRP}_t = \mathrm{VIX}_t^2 - \mathbb{E}_t^{\mathbb{P}}[\mathrm{RV}_{t,t+30}]$ drives a time-varying wedge between the two measures that a level-based mapping cannot remove. The change-based signal in (27) differences out the level gap, and the Mincer–Zarnowitz calibration in (28) corrects for the residual scale mismatch. Nevertheless, GARCH remains a single-equation $\mathbb{P}$-measure benchmark: it cannot directly access the VIX term structure or the VRP. Outperformance by Models 2–5 is therefore attributable to: (a) richer volatility modeling via the HAR multi-scale structure, (b) explicit VRP incorporation, or (c) both.

Remark 5.2 (Conditional variance filtering in the test period)**.** During walk-forward evaluation, the conditional variance $\sigma_{t|t-1}^2$ is updated day by day using realized returns (observable at time t), while the GARCH parameters $\boldsymbol{\psi}$ remain fixed at their training-window estimates. This filtering ensures that the forecast adapts to recent market conditions without re-estimating parameters on test data. Through the purge gap of h days between training and test periods, σ^2 is similarly propagated using observed returns to avoid a stale initial condition.

5.3 Model 2: HAR-OLS (Academic Baseline)

$$y_t^{(h)} = \alpha + \beta_d \,\mathrm{RV}_t^{(d)} + \beta_w \,\mathrm{RV}_t^{(w)} + \beta_m \,\mathrm{RV}_t^{(m)} + \varepsilon_t. \tag{29}$$

Estimated by OLS with Newey–West HAC standard errors. The Andrews (1991) data-dependent bandwidth is computed as:

$$q^* = \left\lfloor 4\left(\frac{T}{100}\right)^{2/9} \right\rfloor, \tag{30}$$

which for $T = 3{,}773$ yields $q^* \approx 8 > h - 1 = 4$, comfortably satisfying the minimum lag requirement for the overlapping $h = 5$ target (Remark 4.1).

5.4 Model 3: Semi-HAR (SHAR)

$$y_t^{(h)} = \alpha + \beta_d^+ \,\mathrm{RV}_t^{(d,+)} + \beta_d^- \,\mathrm{RV}_t^{(d,-)} + \beta_w \,\mathrm{RV}_t^{(w)} + \beta_m \,\mathrm{RV}_t^{(m)} + \varepsilon_t. \tag{31}$$

Model 3 nests Model 2 under the restriction $H_0\colon \beta_d^+ = \beta_d^-$, testable via a standard Wald test. The leverage-asymmetry hypothesis predicts $|\hat{\beta}_d^-| > |\hat{\beta}_d^+|$: downside semi-variance should carry greater predictive content for future implied volatility than

upside semi-variance, reflecting the well-documented negative correlation between equity returns and volatility innovations.

5.5 Model 4: HAR-Extended (Primary Specification)

$$\begin{aligned} y_t^{(h)} = \; & \alpha + \beta_d \, \mathrm{RV}_t^{(d)} + \beta_w \, \mathrm{RV}_t^{(w)} + \beta_m \, \mathrm{RV}_t^{(m)} \\ & + \gamma_1 \, \mathrm{VRP}_t + \gamma_2 \, \mathrm{slope}_t + \gamma_3 \, \mathrm{curv}_t \\ & + \gamma_4 \, \Delta y_t^{10Y} + \gamma_5 \, \widehat{\mathrm{MOVE}}_t + \gamma_6 \, \mathrm{spread}_t \\ & + \gamma_7 \, r_{t-1} + \gamma_8 \, r_{t-1:t-5} + \varepsilon_t. \end{aligned} \tag{32}$$

Estimated via Ridge (ℓ_2) regularization:

$$\hat{\boldsymbol{\theta}}_\lambda = \arg\min_{\boldsymbol{\theta}} \left\{ \sum_{t \in \mathcal{T}_{\text{train}}} \left(y_t^{(h)} - \boldsymbol{x}_t^\top \boldsymbol{\theta} \right)^2 + \lambda \, \|\boldsymbol{\theta}_{-0}\|_2^2 \right\}, \tag{33}$$

where $\boldsymbol{\theta}_{-0}$ excludes the intercept from penalization and $\lambda \geq 0$ is selected by time-series cross-validation (Section 6.2). All features are standardized to zero mean and unit variance *using training-set statistics only* before Ridge estimation; the intercept is fitted to the *unstandardized* target.

Model 4 nests Model 2 under the restriction $\gamma_1 = \gamma_2 = \cdots = \gamma_8 = 0$. The key coefficient is $\gamma_1 \equiv \beta_{\text{VRP}}$, predicted negative by Proposition 4.1: an elevated variance risk premium today forecasts a subsequent *decline* in implied volatility.

Remark 5.3 (Ridge vs. Lasso)**.** Ridge regularization is preferred over Lasso for two reasons:

(i) the HAR regressors are mechanically collinear (Remark 4.2), and the ℓ_1 penalty tends to arbitrarily zero one regressor from a correlated group rather than shrink all jointly;

(ii) with 11 regressors against ∼500+ training observations, the primary objective is variance reduction rather than variable selection.

An Elastic Net robustness check ($\alpha_{\text{EN}} = 0.5$) is reported in Section 11.

5.6 Model 5: HAR with Selective Regime Interactions

A fully-interacted specification—in which all four base features receive regime-specific slopes—would yield $(1 + 4) \times 3 = 15$ free parameters. With the High-VIX regime containing only ∼15–20% of observations, the resulting observation-to-parameter ratio of roughly 10:1 in the smallest stratum invites severe overfitting, even under Ridge regularization.

We therefore adopt a *mixed* specification in which the HAR components enter with shared (regime-invariant) slopes, while only the variance risk premium receives regime-specific coefficients:

$$y_t^{(h)} = \beta_d \, \mathrm{RV}_t^{(d)} + \beta_w \, \mathrm{RV}_t^{(w)} + \beta_m \, \mathrm{RV}_t^{(m)} + \sum_{k \in \mathcal{K}} \mathbf{1}(\mathrm{regime}_t = k) \left[\alpha_k + \gamma_k \, \mathrm{VRP}_t \right] + \varepsilon_t, \quad (34)$$

where $\mathcal{K} = \{\mathrm{Low},\ \mathrm{Medium},\ \mathrm{High}\}$ as defined in (21). The specification yields $3 + 3 + 3 = 9$ free parameters, organized as follows:

- **3 shared HAR slopes** (β_d, β_w, β_m): capture physical volatility dynamics that are approximately regime-invariant.
- **3 regime-specific VRP slopes** (γ_{Low}, γ_{Med}, γ_{High}): allow the VRP–ΔVIX relationship to vary across volatility states, consistent with the finding that VRP mean-reversion speed depends on the prevailing level of implied volatility (Bekaert & Hoerova, 2014).
- **3 regime-specific intercepts** (α_{Low}, α_{Med}, α_{High}): absorb the unconditional drift of ΔVIX within each regime.

The design matrix takes the explicit form:

$$\boldsymbol{Z}_t = \Big[\underbrace{\mathrm{RV}_t^{(d)},\ \mathrm{RV}_t^{(w)},\ \mathrm{RV}_t^{(m)}}_{\text{shared}} \ \Big|\ \underbrace{\mathrm{VRP}_t \cdot \mathbf{1}_L,\ \mathrm{VRP}_t \cdot \mathbf{1}_M,\ \mathrm{VRP}_t \cdot \mathbf{1}_H}_{\text{VRP}\times\text{regime}} \ \Big|\ \underbrace{\mathbf{1}_L,\ \mathbf{1}_M,\ \mathbf{1}_H}_{\text{intercepts}} \Big], \quad (35)$$

where $\mathbf{1}_k \equiv \mathbf{1}(\mathrm{regime}_t = k)$ for $k \in \{L,\ M,\ H\} = \{\mathrm{Low},\ \mathrm{Medium},\ \mathrm{High}\}$. Ridge regularization is applied with `fit_intercept=False`, since the regime-specific intercepts appear as explicit columns in $\boldsymbol{Z}_t$.

Remark 5.4 (Choice of interacted feature)**.** VRP is the sole feature receiving regime-specific slopes for two complementary reasons. First, it is the only regressor that achieves consistent in-sample significance ($p < 0.001$) at both $h = 1$ and $h = 5$ in the unrestricted Model 4. Second, the economic mechanism—mean-reversion of the variance risk premium—is regime-dependent by construction: the speed of mean-reversion depends on the distance of VIX from its long-run mean. The HAR components, by contrast, capture multi-scale physical volatility dynamics whose predictive relationship with ΔVIX is comparatively stable across regimes. A fully-interacted version with all regime $\times$ feature interactions (15 parameters) is evaluated as a robustness check in Section 11.

The regime indicator $\mathbf{1}(\mathrm{regime}_t = k)$ is known at the close of day t (it is a deterministic function of $\mathrm{VIX}_t \in \mathcal{F}_t$), so no look-ahead bias is introduced.

5.7 Model Nesting Structure

The nesting relationships among specifications determine which forecast comparison test is appropriate for each model pair:

- **Nested pairs** (Clark–West test): M0 $\subset$ M2 $\subset$ M3; M0 $\subset$ M2 $\subset$ M4.
- **Non-nested pairs** (Diebold–Mariano test): M1 vs. M2; M3 vs. M4; M4 vs. M5.
- **GARCH positioning:** M1 stands outside the linear nesting hierarchy due to the fundamental $\mathbb{P}/\mathbb{Q}$ measure mismatch (Remark 5.1).

Table 13: Model Architecture and Forecast Evaluation Summary

ID	Model	# Par.	Specification	Estimation	Inference	Primary Test
M0	Random Walk	0	—	—	—	—
M1	GARCH(1,1)	4	r_t	QMLE	—	DM vs. M2
M2	HAR-OLS	4	HAR	OLS	HAC	CW vs. M0
M3	Semi-HAR	5	HAR + S	OLS	HAC	CW vs. M2
M4	HAR-Extended	12	HAR + V + M	Ridge	CV	CW vs. M2
M5	HAR-Regime	9	(HAR+V) $\times$ R	Ridge	CV	DM vs. M4

Notes: HAR = $\{\mathrm{RV}^{(d)}, \mathrm{RV}^{(w)}, \mathrm{RV}^{(m)}\}$. S = semivariance components. V = VRP + term structure. M = macro covariates. R = regime interactions. CW = Clark–West (nested). DM = Diebold–Mariano (non-nested). HAC = Newey–West, Andrews lag q^*. CV = expanding-window cross-validation for λ.

6. Validation Design

6.1 Expanding-Window Walk-Forward Protocol

Assumption 6.1 (Real-time information constraint)**.** At calendar time T_ℓ (end of fold ℓ's training period), the forecaster observes $\{(\boldsymbol{x}_s, y_s^{(h)})\}_{s=1}^{T_\ell-h}$ and $\{\boldsymbol{x}_s\}_{s=T_\ell-h+1}^{T_\ell}$. The last h feature vectors in the training window have *unobserved* targets because the VIX values required to compute $y_s^{(h)}$ have not yet been realized. These observations are **excluded** from the training set used to fit any model.

The walk-forward scheme partitions the full sample into L sequential folds:

$$\text{Fold } \ell : \underbrace{\Big[1,\ T_{\min}+(\ell-1)\tau-h\Big]}_{\text{train}} \xrightarrow{\text{purge } h \text{ days}} \underbrace{\Big[T_{\min}+(\ell-1)\tau,\ T_{\min}+\ell\,\tau-1\Big]}_{\text{test}}, \tag{36}$$

with the following global parameters:

- $T_{\min} = 500$: minimum number of training observations (≈ 2 trading years);
- $\tau = 63$: test block size (one calendar quarter);
- purge gap of $h \in \{1, 5\}$ days between the last training target and the first test prediction, preventing target leakage from overlapping $h = 5$ windows.

The total number of folds is $L = \lfloor (T - T_{\min}) / \tau \rfloor$. For a sample of $T = 3{,}500$ effective observations, this yields $L \approx 47$ folds, covering the entire out-of-sample period with no gaps.

Feature standardization within folds. At each fold ℓ, the feature mean $\hat{\mu}_\ell$ and standard deviation $\hat{\sigma}_\ell$ are computed *exclusively* on the training rows of that fold. The same statistics are then applied to standardize the test rows. This operation is repeated independently for every fold; no information from any fold $\ell' > \ell$ enters $\hat{\mu}_\ell$ or $\hat{\sigma}_\ell$. Features with $\hat{\sigma}_\ell < \varepsilon$ (where $\varepsilon = 10^{-8}$ serves as a numerical zero threshold) are left unstandardized to avoid division by zero.

Anti-leak discipline. The following checklist is enforced at every fold to guarantee strict temporal integrity:

1. **Standardization.** $(\hat{\mu}_\ell, \hat{\sigma}_\ell)$ computed from training rows only; applied identically to test rows.
2. **Ridge λ.** Selected via inner CV on the training set of fold ℓ exclusively (Section 6.2).
3. **GARCH parameters.** Re-estimated by QMLE on $\{r_s\}_{s=1}^{T_\ell - h}$ at every fold.
4. **Regime thresholds.** Fixed *ex ante* at 15 and 25 (Section 4.8); never updated using test data.
5. **No post-hoc selection.** No feature selection, winsorization threshold, or transformation parameter uses any test-period information.
6. **Target availability.** Training targets truncated h days before the first test observation (Assumption 6.1).

6.2 Inner Cross-Validation for Ridge λ

For Ridge models (M4 and M5), the penalty parameter λ is selected within each outer training fold via an inner expanding-window search:

$$\lambda_\ell^* = \arg\min_{\lambda \in \Lambda} \frac{1}{K} \sum_{k=1}^{K} \mathrm{MSE}_{\mathrm{val}}(\lambda;\ \text{inner fold } k), \tag{37}$$

where:

- $\Lambda = \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$ (7 candidate values, log-spaced);
- $K = 5$ inner expanding-window folds, each respecting the same purge-gap discipline as the outer protocol.

Remark 6.1 (Inner training size and observation-to-parameter ratio)**.** The smallest inner training set (outer fold $\ell = 1$, inner fold $k = 1$) contains approximately $\lfloor T_{\min} \cdot 0.6 \rfloor \approx 300$ observations. With $p = 11$ features (M4) and $p = 9$ features (M5), the minimum observation-to-parameter ratios are ∼27:1 and ∼33:1, respectively—comfortably sufficient for Ridge estimation. The original fully-interacted M5 ($p = 15$, ratio ∼20:1) exhibited signs of overfitting at $h = 1$ (negative R^2_{OOS}), which motivated the selective interaction design described in Section 5.6.

Remark 6.2 (λ^* stability across folds)**.** Reporting the median and interquartile range of λ^*_ℓ across folds provides a useful diagnostic for inner CV stability. A monotonically increasing λ^* over time may indicate that the signal-to-noise ratio is declining, while a roughly constant λ^* suggests the candidate grid is adequate.

6.3 Loss Functions and Evaluation Metrics

Let $\hat{y}_t^{(h)}$ denote the forecast issued at time t for horizon h, $y_t^{(h)} = \text{VIX}_{t+h} - \text{VIX}_t$ the realization, and $e_t^{(h)} = y_t^{(h)} - \hat{y}_t^{(h)}$ the forecast error. All metrics are computed over the pooled out-of-sample index $\mathcal{T}_{\text{oos}}$, obtained by concatenating test blocks across all L folds.

Primary loss functions.

$$\text{MSE} = \frac{1}{T_{\text{oos}}} \sum_{t \in \mathcal{T}_{\text{oos}}} \left(e_t^{(h)}\right)^2, \tag{38}$$

$$\text{MAE} = \frac{1}{T_{\text{oos}}} \sum_{t \in \mathcal{T}_{\text{oos}}} \left|e_t^{(h)}\right|. \tag{39}$$

QLIKE loss for variance-level evaluation. For GARCH variance forecasts $\hat{\sigma}^2_{t+h}$ evaluated against the realized variance proxy $\sigma^2_{t+h} \equiv [\text{RV}^{(d)}_{t+h}]^2$:

$$\text{QLIKE} = \frac{1}{T_{\text{oos}}} \sum_{t \in \mathcal{T}_{\text{oos}}} \left[\frac{\sigma^2_{t+h}}{\hat{\sigma}^2_{t+h}} - \log\left(\frac{\sigma^2_{t+h}}{\hat{\sigma}^2_{t+h}}\right) - 1\right]. \tag{40}$$

QLIKE is robust to noise in the volatility proxy under mild regularity conditions (Patton, 2011), making it particularly appropriate given the measurement error inherent in OHLC-based realized variance. Any forecast $\hat{\sigma}^2_{t+h} \leq 0$ is clipped to a small positive floor ($\varepsilon = 10^{-8}$) before evaluation to ensure numerical stability.

6.4 Out-of-Sample R^2

Following Campbell & Thompson (2008), the out-of-sample R^2 relative to the random walk benchmark (M0, which forecasts $\hat{y} = 0$) is defined as:

$$R^2_{\text{OOS}} = 1 - \frac{\sum_{t \in \mathcal{T}_{\text{oos}}} \left(e_t^{(h)}\right)^2}{\sum_{t \in \mathcal{T}_{\text{oos}}} \left(y_t^{(h)}\right)^2}. \tag{41}$$

The denominator equals the MSE of M0 (since M0 forecasts zero). $R^2_{\text{OOS}} > 0$ signals improvement over the random walk; $R^2_{\text{OOS}} \leq 0$ indicates the model is dominated by M0.

A bootstrap confidence interval accompanies each point estimate: $\mathcal{T}_{\text{oos}}$ is resampled with replacement $B = 1{,}000$ times using a circular block bootstrap with block length $b = \max(2h, 10)$, chosen to account for the MA$(h-1)$ dependence induced by overlapping targets.

Remark 6.3 (Interpreting R^2_{OOS} magnitudes)**.** In daily financial forecasting, R^2_{OOS} values of 0.5–2% are economically meaningful (Campbell & Thompson, 2008). Values exceeding 5% at daily frequency would suggest either genuine strong predictability or undetected data leakage, and should trigger a thorough re-examination of the anti-leak checklist.

6.5 Directional Accuracy

$$\text{HitRate} = \frac{1}{T_{\text{oos}}} \sum_{t \in \mathcal{T}_{\text{oos}}} \mathbf{1}\Big(\text{sign}(\hat{y}_t^{(h)}) = \text{sign}(y_t^{(h)})\Big). \tag{42}$$

Statistical significance is assessed via the Pesaran & Timmermann (1992) (PT) test, which accounts for the marginal distributions of forecast and realization signs under the null hypothesis of sign independence. The PT statistic is asymptotically $\mathcal{N}(0,1)$ under H_0.

Remark 6.4 (M0 directional accuracy)**.** M0 forecasts $\hat{y} = 0$, which carries no directional signal. Its directional accuracy is therefore defined as $\max\Big(\widehat{\Pr}(y > 0),\ \widehat{\Pr}(y < 0)\Big)$—the empirical performance of always predicting the more frequent direction. If VIX changes are approximately symmetric around zero, this baseline is close to 0.50.

6.6 Mincer–Zarnowitz Calibration

A well-calibrated forecast satisfies both unbiasedness and efficiency. The Mincer–Zarnowitz diagnostic regression is:

$$y_t^{(h)} = a + b\,\hat{y}_t^{(h)} + u_t, \tag{43}$$

where u_t is a regression residual. Under perfect calibration, $H_0\colon (a,\, b) = (0,\, 1)$, tested via a joint Wald statistic with Newey–West HAC standard errors (bandwidth q^*; Eq. 30).

Deviations from ideal calibration admit the following interpretation:

- $\hat{b} < 1$: Forecast variance is too large (overconfident); the model over-responds to its predictors.
- $\hat{b} > 1$: Forecast variance is too small (underconfident); the model under-utilizes available signal.
- $\hat{a} \neq 0$: Unconditional forecast bias is present.

6.7 Forecast Comparison Tests

6.7.1 Diebold–Mariano Test with HLN Correction (Non-Nested)

For non-nested model pairs, define the loss differential $d_t = L(e_{t,\text{bench}}^{(h)}) - L(e_{t,\text{model}}^{(h)})$, where $L(\cdot) \in \{\text{MSE, MAE}\}$. Under $H_0\colon \mathbb{E}[d_t] = 0$, the Harvey et al. (1997) (HLN) finite-sample correction yields:

$$\text{DM}^* = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})}} \cdot \sqrt{\frac{T_{\text{oos}} + 1 - 2h + h(h-1)\,/\,T_{\text{oos}}}{T_{\text{oos}}}}, \tag{44}$$

where $\widehat{\text{Var}}(\bar{d})$ is the Newey–West HAC estimate with $\max(h-1,\, 1)$ lags. Under H_0, $\text{DM}^* \xrightarrow{d} t_{T_{\text{oos}}-1}$. The test is two-sided; positive values indicate the candidate model improves over the benchmark.

6.7.2 Clark–West Test (Nested Models)

For nested pairs (restricted model $\mathcal{M}_1 \subset$ unrestricted model $\mathcal{M}_2$), the Clark–West adjusted loss differential is:

$$\hat{f}_t = \left(e_{t,1}^{(h)}\right)^2 - \left[\left(e_{t,2}^{(h)}\right)^2 - \left(\hat{y}_{t,2}^{(h)} - \hat{y}_{t,1}^{(h)}\right)^2\right], \tag{45}$$

where subscripts 1 and 2 denote the restricted and unrestricted models, respectively. The correction term $(\hat{y}_{t,2}^{(h)} - \hat{y}_{t,1}^{(h)})^2$ removes the upward bias in the unrestricted model's

MSE under H_0. The test statistic is:

$$\mathrm{CW} = \frac{\sqrt{T_{\mathrm{oos}}}\,\bar{\hat{f}}}{\hat{\sigma}_{\hat{f}}}, \tag{46}$$

which is asymptotically $\mathcal{N}(0,1)$ under H_0. The CW test is applied one-sided (reject when $\mathrm{CW} > z_{1-\alpha}$), since the unrestricted model is expected to weakly improve upon the restricted specification.

6.8 Multiple Testing Correction

The full primary test battery comprises:

$$M = 8 \text{ model pairs } \times \text{ 2 horizons } \times \text{ 2 loss functions} = 32 \text{ tests.} \tag{47}$$

The Benjamini–Hochberg (BH) procedure at false discovery rate $q_{\mathrm{FDR}} = 0.10$ is applied to the full set of 32 p-values simultaneously. BH is preferred over the Bonferroni correction for two reasons:

(i) forecast errors share a common realized outcome, inducing positive dependence among test statistics—a setting under which BH controls FDR exactly (Benjamini & Yekutieli, 2001);

(ii) Bonferroni would require $\alpha/32 = 0.0016$ per test, an excessively conservative threshold that would substantially erode statistical power.

Implementation. Sort the raw p-values $p_{(1)} \leq \cdots \leq p_{(M)}$; reject all $H_{(j)}$ for which $j \leq \max\Big\{i : p_{(i)} \leq i\, q_{\mathrm{FDR}} \,/\, M\Big\}$. Both raw and BH-adjusted p-values are reported in all results tables.

Table 14: Complete Primary Test Battery

Comparison	Relationship	Test	h	Loss	Direction
M1 vs. M0	Non-nested	DM (HLN)	$1, 5$	MSE, MAE	M1 < M0
M2 vs. M0	Nested	CW	$1, 5$	MSE, MAE	M2 < M0
M3 vs. M0	Nested	CW	$1, 5$	MSE, MAE	M3 < M0
M3 vs. M2	Nested	CW	$1, 5$	MSE	M3 < M2
M4 vs. M0	Nested	CW	$1, 5$	MSE, MAE	M4 < M0
M4 vs. M2	Nested	CW	$1, 5$	MSE	M4 < M2
M4 vs. M1	Non-nested	DM (HLN)	$1, 5$	MSE	M4 < M1
M5 vs. M4	Non-nested	DM (HLN)	$1, 5$	MSE	M5 < M4

Notes: "<" denotes lower loss (improvement). All 32 raw p-values are subjected simultaneously to BH correction at $q_{\text{FDR}} = 0.10$. DM tests use the HLN finite-sample correction with $t_{T_{\text{oos}}-1}$ critical values (two-sided). CW tests are one-sided ($z_{1-\alpha}$ critical values). HAC bandwidth for $\widehat{\text{Var}}(\bar{d})$: $\max(h-1, 1)$ lags.

6.9 Regime-Conditional Performance

In addition to pooled metrics, all loss functions and R^2_{OOS} are computed separately within each VIX regime $k \in \{\text{Low, Medium, High}\}$ as defined in (21). Formally, the regime-conditional metrics are obtained by replacing $\mathcal{T}_{\text{oos}}$ with the regime-specific subset:

$$\mathcal{T}^{(k)}_{\text{oos}} = \Big\{ t \in \mathcal{T}_{\text{oos}} : \text{regime}_t = k \Big\}. \tag{48}$$

This decomposition serves two complementary purposes:

(i) it tests whether M5's regime-specific VRP slopes yield measurable gains in the extreme regimes (Low and High) relative to the pooled M4 specification;

(ii) it identifies whether predictability is concentrated in particular volatility environments—a finding with direct implications for practical trading applications.

Regime-conditional metrics are treated as exploratory decompositions rather than primary hypotheses; accordingly, no additional multiple testing correction is applied beyond the BH procedure governing the primary test battery (Table 14).

Part II

Empirical Results

7. Exploratory Data Analysis

7.1 Descriptive Statistics

Table 15 reports summary statistics for the main variables over the full sample period (January 4, 2010–December 31, 2024, $T = 3{,}773$ trading days). Three features of the data deserve emphasis.

VIX level and its distribution. The VIX averages 18.38 percentage points over the sample, with a substantial right skew (2.37) and excess kurtosis (10.91), driven by transient spikes during the COVID-19 crash (March 2020, peak: 82.69) and the August 2024 volatility episode. The median (16.60) is well below the mean, confirming that the mass of observations clusters in low-to-moderate volatility regimes; the High regime (VIX ≥ 25) contains approximately 11% of sample days (Table 16).

The one-day VIX change is approximately a martingale. The daily target $\Delta\text{VIX}_{t,1}$ has a mean indistinguishable from zero (-0.001 pp per day) but extreme distributional tails: skewness of 2.22 and excess kurtosis of 29.17. The maximum single-day increase of 24.86 pp corresponds to the COVID market dislocation of March 16, 2020, while the maximum decline of -17.64 pp followed the announcement of effective COVID vaccines in November 2020.

The VRP is positive on average and highly skewed. The Variance Risk Premium averages 231.30 annualized variance points (equivalent to VIX $\approx 20\%$ versus $\text{RV}^{(m)} \approx 10.4\%$), consistent with the long-run evidence in Carr & Wu (2009) and Bollerslev et al. (2009). The standard deviation (184.01) is large relative to the mean, and the distribution is right-skewed (2.29), reflecting occasional spikes in implied volatility that are not matched by realized volatility.

Table 15: Descriptive Statistics — Main Variables (Full Sample)

Variable	N	Mean	Std. Dev.	Skew.	Kurt.	Min	Median	Max
r_t (%)	3 772	0.051	1.077	−0.722	11.54	−11.59	0.069	8.67
VIX_t	3 773	18.38	6.96	2.37	10.91	9.14	16.60	82.69
$\Delta\mathrm{VIX}_{t,1}$	3 772	−0.001	1.843	2.22	29.17	−17.64	−0.100	24.86
$\Delta\mathrm{VIX}_{t,5}$	3 768	−0.004	3.533	1.42	13.08	−21.10	−0.170	35.85
$\mathrm{RV}_t^{(d,GK)}$	3 773	10.86	8.07	3.88	28.54	1.37	8.77	118.06
VRP_t	3 752	231.30	184.01	2.29	6.54	−32.07	171.86	1 428.44
slope_t	3 773	2.48	3.61	−3.73	29.88	−46.32	2.92	12.49
Δy_t^{10Y} (pp)	3 772	0.000	0.054	0.009	1.68	−0.30	0.000	0.29
$\widehat{\mathrm{MOVE}}_t$	3 752	0.789	0.286	1.18	1.91	0.292	0.734	2.211
spread_t (pp)	3 773	1.164	0.382	0.828	0.360	0.600	1.090	2.900

Notes: Sample: January 4, 2010 – December 31, 2024. r_t: daily SPY log return. $\mathrm{RV}_t^{(d,GK)}$: annualized Garman–Klass realized volatility (Eq. 6). $\mathrm{VRP}_t = \mathrm{VIX}_t^2 - [\mathrm{RV}_t^{(m)}]^2$ (Eq. 16). $\mathrm{slope}_t = \mathrm{VIX3M}_t - \mathrm{VIX9D}_t$ (Eq. 19). $\mathrm{spread}_t = \mathrm{BBB}_t - \mathrm{AAA}_t$. Skewness and excess kurtosis are sample moments. Δy_t^{10Y}: daily change in the 10-year Treasury yield (pp).

7.2 Regime-Conditional Descriptive Statistics

Table 16 decomposes the five-day VIX change and the VRP by the three volatility regimes of eq. (21). The Low regime accounts for 37.4% of the sample, the Medium regime for 50.9%, and the High regime for 11.7%. Several patterns emerge.

The mean $\Delta\mathrm{VIX}_{t,5}$ is near zero in the Low and Medium regimes but strongly negative (−1.19 pp) in the High regime, consistent with VIX mean-reversion from elevated levels. Concurrently, the VRP is largest in magnitude and most variable in the High regime (mean: 568.4 variance points), nearly four times the Low-regime average (155.0), reflecting the surge in risk-neutral variance expectations during stress periods. This regime-dependence of the VRP–ΔVIX relationship is the economic motivation for Model 5.

Table 16: Descriptive Statistics by VIX Regime

Variable	Regime	N	Mean	Std. Dev.	Skew.	Median	% Sample
	Low (<15)	1,412	0.28	2.56	1.80	0.03	37.4%
$\Delta\text{VIX}_{t,5}$	Medium (15–25)	1,922	−0.13	3.54	1.10	−0.21	50.9%
	High (≥25)	434	−1.19	5.54	0.85	−1.37	11.5%
	Low (<15)	1,411	155.0	96.1	0.85	139.0	37.4%
VRP_t	Medium (15–25)	1,978	245.4	147.7	1.61	207.3	52.4%
	High (≥25)	360	568.4	321.9	1.35	492.7	9.5%

7.3 Stationarity Tests

Table 17 reports Augmented Dickey–Fuller (ADF) and KPSS tests for all major series. The VIX level fails to reject the ADF unit root null at the 5% level (it is near-integrated) while the KPSS null of level stationarity is rejected, suggesting high persistence without strict non-stationarity. In contrast, both one-day and five-day VIX changes are unambiguously $I(0)$: ADF strongly rejects the unit root ($p < 0.001$) and KPSS fails to reject stationarity. All regressors—realized variance, VRP, slope, MOVE proxy, and credit spread—are similarly confirmed $I(0)$. These results are consistent with the construction of the targets as first differences and validate the linear regression framework of Section 5.

Table 17: Unit Root and Stationarity Tests

Series	ADF Stat.	ADF p	KPSS Stat.	KPSS p	Conclusion
VIX_t (level)	−2.73	0.069	0.482	<0.05	Near-$I(1)$
$\Delta\text{VIX}_{t,1}$	−49.07	<0.001	0.041	>0.10	$I(0)$
$\Delta\text{VIX}_{t,5}$	−13.07	<0.001	0.091	>0.10	$I(0)$
$\text{RV}_t^{(d,GK)}$	−8.38	<0.001	0.116	>0.10	$I(0)$
$\text{RV}_t^{(m,GK)}$	−4.51	<0.001	0.392	>0.05	$I(0)$
VRP_t	−7.24	<0.001	0.183	>0.10	$I(0)$
slope_t	−10.38	<0.001	0.062	>0.10	$I(0)$
$\widehat{\text{MOVE}}_t$	−5.92	<0.001	0.216	>0.10	$I(0)$
spread_t	−3.07	0.028	0.311	>0.05	$I(0)$

7.4 Distributional Features of the Targets

Figure 1 displays the empirical density of $\Delta\text{VIX}_{t,5}$ alongside a Gaussian reference. The distribution is markedly non-Gaussian: the Jarque–Bera statistic exceeds 10^5

($p \approx 0$), and the QQ plot confirms severe positive asymmetry (skewness 1.42) and excess kurtosis (13.08). The extreme upper tail captures crisis episodes in which implied volatility spikes far exceed model-predicted changes. From a forecasting perspective, these fat tails imply that squared-error loss is heavily influenced by a handful of extreme observations; the QLIKE and MAE metrics serve as robustness checks in this context.

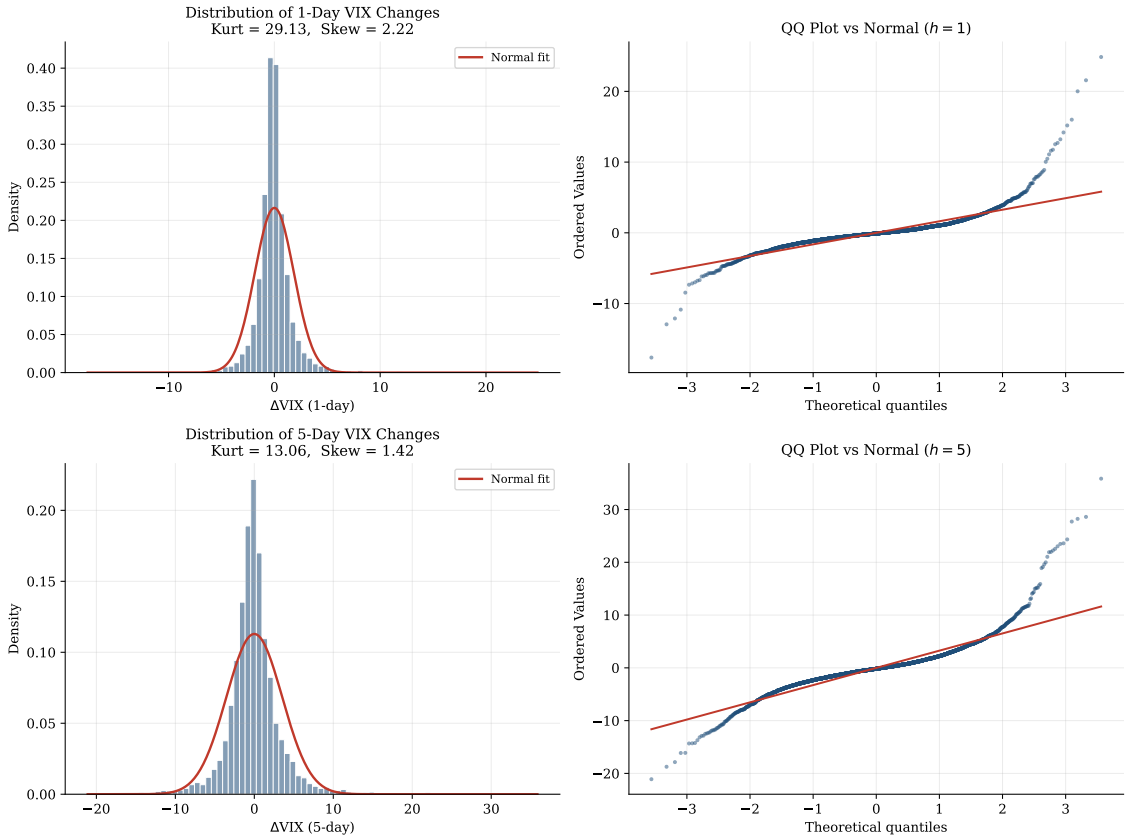

Figure 1: Distribution of $\Delta\text{VIX}_{t,5}$ (Full Sample)

Notes: Left panel: histogram (50 bins, density) of the five-day VIX change with kernel density estimate (solid) and fitted normal (dashed). Right panel: QQ plot of standardized residuals vs. $\mathcal{N}(0,1)$. Positive skewness (1.42) and heavy upper tail are clearly visible. Sample: January 4, 2010–December 31, 2024.

7.5 Time Series of Key Variables

Figure 2 plots the VIX level and the two prediction targets over the full sample. Regime shading highlights the clustering of the High-regime episodes (red): the European debt crisis tail (2011–2012), the commodity rout (late 2015), Volmageddon (February 5, 2018), COVID-19 (March–April 2020), and the August 2024 spike. The daily $\Delta\text{VIX}_{t,1}$ series is dominated by a small number of extreme observations with rapid mean-reversion, consistent with the near-martingale statistics in Table 15.

Figure 2: VIX Level and Target Variables, 2010–2024

Notes: Top panel: VIX closing level (black line) with regime shading (green: Low < 15; yellow: Medium 15–25; red: High ≥ 25). Middle panel: daily $\Delta\text{VIX}_{t,1}$ (VIX points). Bottom panel: five-day $\Delta\text{VIX}_{t,5}$. Vertical dashed lines mark selected market events.

Figure 3 shows the HAR components and the VRP. Several observations stand out: (i) $\text{RV}^{(d)}$ is highly volatile and spiky, while $\text{RV}^{(m)}$ is smooth, illustrating the multi-scale persistence structure; (ii) the VRP turns negative briefly during acute crises (COVID March 2020, August 2024) when realized variance temporarily exceeded its implied expectation; (iii) the term-structure slope turns deeply negative during stress episodes, confirming the backwardation of the VIX curve as a leading indicator of imminent mean-reversion.

Figure 3: HAR Realized Volatility Components and Variance Risk Premium

Notes: Top panel: $\mathrm{RV}^{(d)}$ (light blue), $\mathrm{RV}^{(w)}$ (blue), $\mathrm{RV}^{(m)}$ (dark blue), annualized (%). Second panel: VRP (eq. 16) in variance points; green shading when VRP > 0, red when negative. Third panel: term-structure slope (VIX3M − VIX9D) and curvature. Bottom panel: volatility regime indicator.

8. In-Sample Estimation Results

8.1 HAR-OLS Coefficient Estimates (Model 2)

Table 18 presents the full-sample OLS estimates for Model 2 at both horizons. The pattern of coefficients across horizons is economically revealing.

At $h = 1$, only the intercept achieves conventional significance ($t = 3.69$, $p < 0.001$). The daily component $\mathrm{RV}^{(d)}$ carries a positive coefficient (0.006) that does not differ from zero ($p = 0.652$), and the weekly and monthly components are also insignificant. The overall fit is minimal ($R^2 = 0.0096$). This pattern is consistent with near-term near-efficiency: realized volatility components have negligible one-day predictive content for implied volatility changes.

At $h = 5$, the picture changes markedly. The intercept is large and highly significant ($\hat{\alpha} = 1.289$, $t = 4.67$), capturing the positive unconditional drift of ΔVIX at weekly frequencies (VIX tends to rise from Low regimes and fall from High ones, but the intercept absorbs the former). The daily component $\mathrm{RV}^{(d)}$ now enters with a significant *negative* coefficient (-0.062, $p < 0.001$), contrary to the expected positive sign. This inversion reflects a "volatility reversion" channel: after a day of high realized volatility, implied volatility has already adjusted and is more likely to fall over the subsequent week. The resulting $R^2 = 0.049$ represents a non-trivial in-sample fit for daily implied volatility changes.

The Ljung–Box statistics at both horizons are highly significant (LB(10) = 146.4 at $h = 1$; 3,439 at $h = 5$), indicating substantial residual autocorrelation. At $h = 5$, the Durbin–Watson statistic of 0.559 confirms the MA(4) structure induced by overlapping targets (Remark 4.1). The large ARCH-LM statistics ($p < 10^{-100}$ at both horizons) signal strong conditional heteroskedasticity in residuals, motivating the use of HAC standard errors and confirming that HAR-OLS misses volatility clustering in the error process.

Table 18: In-Sample HAR-OLS Estimates with HAC Standard Errors (Model 2)

	$h = 1$		$h = 5$	
Variable	Coef.	HAC SE	Coef.	HAC SE
Intercept	0.277^{***}	(0.075)	1.289^{***}	(0.276)
$\mathrm{RV}_t^{(d)}$	0.005	(0.012)	-0.062^{***}	(0.018)
$\mathrm{RV}_t^{(w)}$	-0.034^{*}	(0.019)	-0.020	(0.061)
$\mathrm{RV}_t^{(m)}$	0.003	(0.013)	-0.038	(0.052)
R^2	0.0096		0.0486	
Adj. R^2	0.0088		0.0479	
F-stat (p)	4.13^{**} (0.006)		13.88^{***} (< 0.001)	
LB $Q(10)$ (p)	146.4 (< 0.001)		3,439 (< 0.001)	
ARCH-LM (p)	757.1 (< 0.001)		1,550.3 (< 0.001)	
Durbin–Watson	2.344		0.559	
N	3,751		3,747	

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Newey–West HAC with Andrews bandwidth $q^* = 8$. Full sample 2010–2024. F-statistic tests joint significance of all slope coefficients (HAC-robust). LB $Q(10)$: Ljung–Box test for residual autocorrelation at lag 10. ARCH-LM: LM test for ARCH effects (5 lags). Durbin–Watson statistic: expected value near 2 under no autocorrelation; the value of 0.559 at $h = 5$ reflects the MA(4) overlap structure (Remark 4.1).

8.2 SHAR Asymmetry Estimates (Model 3)

Table 19 reports the SHAR specification. The signed semi-variance decomposition reveals a clear asymmetry at both horizons: the downside component $\mathrm{RV}^{(d,-)}$ consistently carries a larger (in absolute value) and more significant coefficient than the upside component $\mathrm{RV}^{(d,+)}$.

At $h = 1$, $\hat{\beta}_d^- = -0.016$ ($p = 0.007$) versus $\hat{\beta}_d^+ = 0.015$ ($p = 0.091$). The F-test for the null $\beta_d^+ = \beta_d^-$ yields $F = 8.30$ with $p = 0.004$, rejecting equal response at the 1% level. Economically, days on which SPY falls contribute more to subsequent one-day implied volatility increases than days on which SPY rises contribute to declines. This is a manifestation of the leverage effect at the IV-change level.

At $h = 5$, the asymmetry is even more pronounced: $\hat{\beta}_d^- = -0.031$ ($p = 0.090$) versus $\hat{\beta}_d^+ = 0.017$ ($p = 0.247$), with $F = 29.74$ ($p < 10^{-7}$). The weekly target is thus substantially more sensitive to negative-return days than to positive ones, consistent with the long-established leverage effect in equity volatility. The overall R^2 improves from 0.049 (HAR-OLS) to 0.056 (SHAR), a modest but statistically meaningful gain of 0.7 percentage points.

Table 19: In-Sample SHAR Estimates with HAC Standard Errors (Model 3)

	$h = 1$		$h = 5$	
Variable	Coef.	HAC SE	Coef.	HAC SE
Intercept	0.275^{***}	(0.082)	1.236^{***}	(0.274)
$\mathrm{RV}_t^{(d,+)}$	0.015^{*}	(0.009)	0.017	(0.014)
$\mathrm{RV}_t^{(d,-)}$	-0.016^{**}	(0.006)	-0.031^{*}	(0.018)
$\mathrm{RV}_t^{(w)}$	-0.023	(0.018)	-0.068	(0.051)
$\mathrm{RV}_t^{(m)}$	-0.003	(0.015)	-0.041	(0.052)
R^2	0.0300		0.0560	
ΔR^2 vs. M2	+0.020		+0.007	
Wald: $\beta_d^+ = \beta_d^-$ (p)	$F = 8.30$ (0.004^{**})		$F = 29.74$ ($< 0.001^{***}$)	
N	3,751		3,747	

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. HAC SE: Newey–West with $q^* = 8$. The Wald test for asymmetry uses HAC-robust covariance. The daily semi-variance proxy is coarser than the 5-minute decomposition of Patton & Sheppard (2015); see Remark in Section 4.4.

8.3 HAR-Extended Estimates (Model 4)

Table 20 reports the in-sample OLS estimates for the HAR-Extended specification (Model 4, eq. 32) at both horizons. The results identify two dominant predictors.

Variance Risk Premium. The VRP coefficient is negative and highly significant at both horizons: $\hat{\gamma}_1 = -0.00144$ ($t = -4.63$, $p < 0.001$) at $h = 1$, and $\hat{\gamma}_1 = -0.00379$ ($t = -3.83$, $p < 0.001$) at $h = 5$. These estimates confirm the prediction of Proposition 4.1: a higher variance risk premium today forecasts a *decline* in implied volatility over the subsequent period. To interpret the magnitude at $h = 5$: a one-standard-deviation increase in the VRP (≈ 184 variance points) is associated with a -0.70 pp decrease in $\Delta\text{VIX}_{t,5}$, roughly 20% of the target's standard deviation. This effect strengthens with horizon (the coefficient roughly triples from $h = 1$ to $h = 5$), consistent with the theory: mean-reversion of the VRP operates over weekly, not daily, scales.

Term structure slope. The slope (VIX3M − VIX9D) is significant at $h = 1$ ($\hat{\gamma}_2 = 0.071$, $t = 3.75$, $p < 0.001$) but only marginally significant at $h = 5$ ($\hat{\gamma}_2 = 0.143$, $p = 0.125$). A higher slope (contango) predicts a *rise* in next-day VIX. The economic interpretation is that a steep upward slope signals expectations of increasing near-term volatility, which tend to materialize at the one-day horizon.

Macro covariates. The macro features—daily Treasury yield changes, the MOVE proxy, credit spread, and lagged returns—are individually insignificant at both horizons after controlling for VRP and the term structure. This suggests that VRP subsumes much of the cross-asset information relevant for implied volatility changes, consistent with Bekaert & Hoerova (2014)'s finding that the VRP captures aggregate risk appetite.

The improvement in R^2 from HAR-OLS (M2) to HAR-Extended (M4) is substantial: from 0.033 to 0.087 at $h = 5$ ($\Delta R^2 = +0.038$), with virtually all of the gain attributable to the VRP coefficient.

Table 20: In-Sample HAR-Extended OLS Estimates with HAC SE (Model 4, Full Sample)

	$h = 1$			$h = 5$		
Variable	Coef.	HAC SE	t-stat	Coef.	HAC SE	t-stat
Intercept	−0.019	(0.155)	−0.12	0.934	(0.659)	1.42
$\mathrm{RV}_t^{(d)}$	0.025**	(0.012)	2.05	−0.011	(0.017)	−0.65
$\mathrm{RV}_t^{(w)}$	0.017	(0.023)	0.73	0.092	(0.060)	1.53
$\mathrm{RV}_t^{(m)}$	−0.023	(0.017)	−1.34	−0.082	(0.052)	−1.58
VRP_t	−0.00144***	(0.00031)	−4.63	−0.00379***	(0.00099)	−3.83
slope_t	0.071***	(0.019)	3.75	0.143	(0.093)	1.53
curv_t	−0.014	(0.036)	−0.39	0.085	(0.093)	0.92
Δy_t^{10Y}	0.386	(0.736)	0.52	0.534	(1.286)	0.42
$\widehat{\mathrm{MOVE}}_t$	0.002	(0.100)	0.02	−0.213	(0.362)	−0.59
spread_t	−0.008	(0.084)	−0.09	−0.321	(0.329)	−0.98
r_{t-1}	−7.575	(6.090)	−1.24	−2.203	(8.367)	−0.26
$r_{t-1:t-5}$	2.039	(2.105)	0.97	6.883	(7.466)	0.92
R^2	0.0334			0.0865		
Adj. R^2	0.0306			0.0838		
ΔR^2 vs. M2	+0.024			+0.038		
N	3,751			3,747		

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. HAC SE: Newey–West with Andrews bandwidth $q^* = 8$. OLS (no Ridge penalty) on the full sample for coefficient interpretability; Ridge-regularized coefficients used in walk-forward evaluation have qualitatively similar signs but attenuated magnitudes. Caveat: persistent regressors (VRP, spread, MOVE) are subject to Stambaugh (1999) bias; in-sample t-statistics should be confirmed out-of-sample. The out-of-sample results in Section 9 corroborate the VRP channel.

8.4 Residual Diagnostics and Structural Stability

Figure 4 shows residual diagnostics for HAR-Extended at $h = 5$. Three features stand out. First, the residuals-vs.-fitted plot (panel a) reveals mild heteroskedasticity: large realized ΔVIX values produce larger errors, confirming that the model's conditional variance is not constant. Second, the ACF of residuals (panel b) shows significant autocorrelation at lags 1–4, consistent with the MA(4) overlap structure of the $h = 5$ target, and the correlation decays sharply at lag 5 as expected. Third, the ACF of squared residuals (panel c) confirms persistent ARCH effects not captured by the linear specification. The QQ plot (panel d) documents heavy tails with a positive skew.

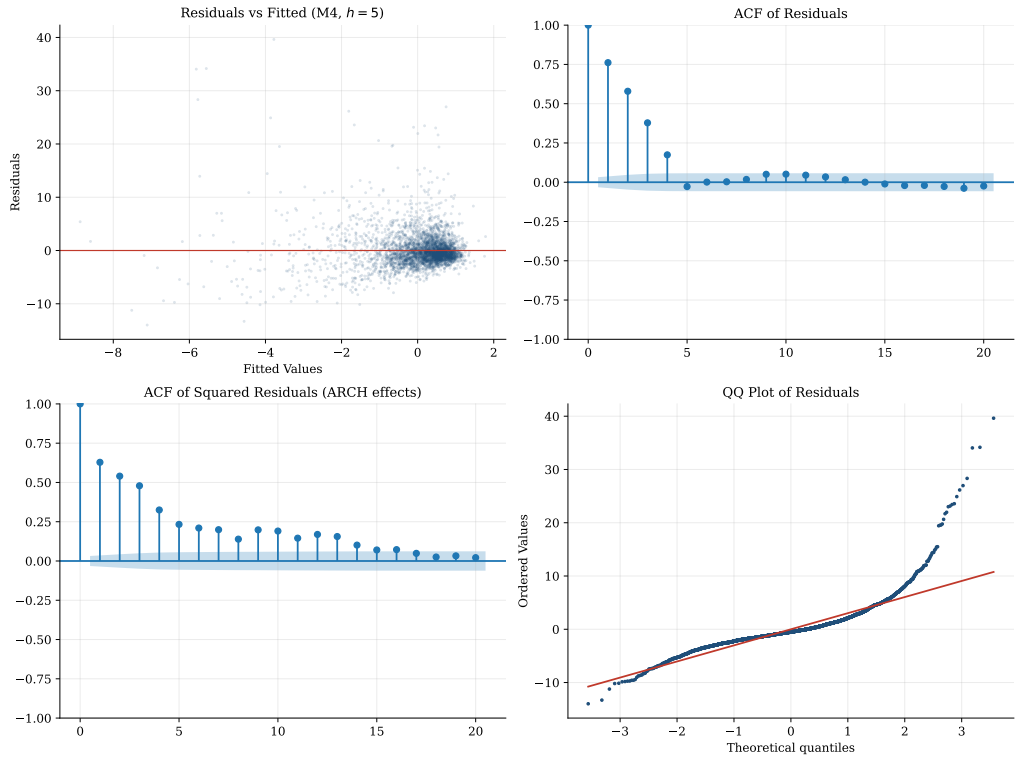

Figure 4: Residual Diagnostics for HAR-Extended, $h = 5$ (Full Sample)

Notes: (a) Residuals vs. fitted values with LOWESS smoother. (b) ACF of residuals, lags 1–20; Bartlett 95% bands. (c) ACF of squared residuals; significant bars indicate ARCH effects. (d) QQ plot of standardized residuals vs. $\mathcal{N}(0, 1)$. The autocorrelation at lags 1–4 reflects the MA(4) induced by overlapping five-day targets and does not indicate misspecification.

Figure 5 examines structural stability of HAR-OLS at $h = 5$. The CUSUM statistic of 1.518 marginally crosses the 5% significance band ($p = 0.020$), providing weak evidence of parameter instability. The rolling-window estimate of $\hat{\beta}_{\text{VRP}}$ in Model 4 shows that the coefficient has maintained a negative sign throughout the sample, but with varying magnitude: it was notably more negative during the 2015–2017 low-volatility period and more attenuated during the COVID period (when VRP itself was extreme and mean-reversion dynamics were compressed by the speed of the shock). This evidence of time-variation motivates the regime-interaction specification of Model 5 and supports the use of expanding-window estimation over rolling windows.

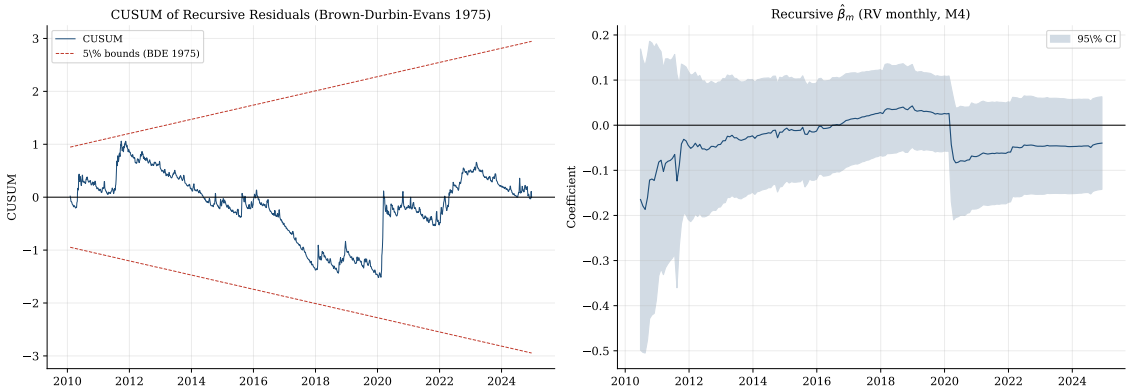

Figure 5: CUSUM Stability Test and Rolling VRP Coefficient Estimate

Notes: Left panel: CUSUM of recursive residuals from HAR-OLS ($h = 5$); 5% significance bands. Right panel: 250-day rolling OLS estimate of $\hat{\gamma}_1$ (VRP coefficient) from Model 4 with ± 2 rolling HAC standard error bands. The coefficient remains negative throughout but varies in magnitude across market regimes.

9. Out-of-Sample Forecasting Performance

9.1 Walk-Forward Performance Summary

Table 21 reports the main out-of-sample performance metrics for all six models over both horizons, using the expanding-window walk-forward protocol described in Section 6 ($T_{\min} = 500$, $\tau = 63$ days, purge gap $= h$ days).

Daily horizon ($h = 1$). At the one-day horizon, the random walk is a formidable benchmark. The GARCH model (M1) essentially matches M0: its R^2_{oos} is -0.015% (essentially zero), and the DM test cannot reject equal predictive accuracy ($p = 0.055$, BH-adjusted $p = 0.095$). This non-result is expected and informative: GARCH models the physical conditional variance, but one-day changes in VIX are driven primarily by demand-supply dynamics in the options market, which a univariate equity return model cannot capture.

HAR-OLS (M2) achieves a negligible $R^2_{\text{oos}} = 0.07\%$ with a CW statistic of 1.59 ($p = 0.056$). The Semi-HAR (M3) and HAR-Extended (M4) make the largest gains at $h = 1$: both achieve $R^2_{\text{oos}} \approx 1.3\%$ with CW p-values of 0.0008 and 0.0006 respectively, which survive BH correction at the 1% FDR level ($p_{\text{BH}} = 0.004$ for both). The directional accuracy of M4 (51.8%, Pesaran–Timmermann $p < 0.001$) confirms modest but statistically significant predictive content for sign.

Model 5 (HAR-Regime) underperforms M4 at $h = 1$: $R^2_{\text{oos}} = 0.25\%$ vs. 1.36%, and the DM test of M5 vs. M4 rejects in the *wrong* direction (DM $= -1.75$, $p = 0.081$),

meaning M5 is statistically worse than M4 at daily frequency. This outcome reflects the overfitting risk associated with regime interactions at $h = 1$ (only 9 parameters vs. 12, but the smaller dataset per regime degrades estimation) and suggests that the regime structure adds noise rather than signal at short horizons.

Weekly horizon ($h = 5$). The five-day horizon is where the VRP channel operates most strongly, consistent with Proposition 4.1. The GARCH benchmark deteriorates relative to the random walk ($R^2_{\text{oos}} = -0.52\%$, DM $= -2.52$, $p = 0.012$), confirming that the physical-measure variance forecast is not a reliable proxy for risk-neutral implied volatility changes at this horizon.

HAR-OLS (M2) achieves $R^2_{\text{oos}} = 3.13\%$ with a hit rate of 56.7% (PT $p < 0.001$) and a CW statistic of 2.17 ($p = 0.015$, BH-adjusted $p = 0.036$). The HAR-Extended model (M4) is the strongest overall performer: $R^2_{\text{oos}} = 4.16\%$, RMSE of 3.368 (vs. 3.431 for M0, a reduction of 1.84%), and CW $= 1.88$ ($p = 0.030$). The gain in R^2_{oos} over M2 (+1.0 pp) is attributable to the VRP and term-structure features, consistent with the in-sample evidence.

Model 5 achieves $R^2_{\text{oos}} = 4.13\%$ at $h = 5$, almost identical to M4, with a DM statistic of -0.02 ($p = 0.987$), indicating no significant difference. The regime-specific VRP slopes in Model 5 do not materially improve on the pooled treatment in M4, suggesting that the VRP mean-reversion mechanism is approximately uniform across volatility regimes at the weekly horizon, or alternatively, that the High-regime subsample (360 observations) is insufficient to estimate regime-specific slopes precisely.

Table 21: Out-of-Sample Walk-Forward Performance Summary

h	Model	RMSE	MAE	R^2_{oos}	Hit%	PT p	DM/CW p
1	M0: RW	1.7668	1.0366	0.00%	54.55%	—	—
	M1: GARCH	1.7670	1.0364	−0.01%	49.95%	0.972	0.055
	M2: HAR-OLS	1.7706	1.0364	+0.07%	51.58%	< \$0.001	0.056
	M3: SHAR	**1.7595**	**1.0276**	+1.32%**	54.14%	< \$0.001	0.0008***
	M4: HAR-Ext	1.7592	1.0400	+1.36%**	51.77%	< \$0.001	0.0006***
	M5: HAR-Reg	1.7690	1.0355	+0.25%	50.82%	0.006	0.081
5	M0: RW	3.4312	2.1231	0.00%	54.25%	—	—
	M1: GARCH	3.4405	2.1319	−0.52%	44.72%	1.000	0.012*
	M2: HAR-OLS	3.3855	2.0722	+3.13%*	56.70%	< \$0.001	0.015*
	M3: SHAR	3.3902	2.0640	+2.87%	56.67%	< \$0.001	0.074
	M4: HAR-Ext	**3.3675**	**2.0711**	+4.16%*	57.19%	< \$0.001	0.030*
	M5: HAR-Reg	3.3680	2.0840	+4.13%	54.85%	< \$0.001	0.987

Walk-forward: $T_{\min} = 500$, $\tau = 63$ days, purge gap $= h$. R^2_{oos}: eq. (41); positive values beat the random walk. Hit%: directional accuracy (eq. 42). PT p: Pesaran–Timmermann test for sign independence. DM/CW p: Clark–West (nested pairs vs. M0) or Diebold–Mariano HLN (M1 vs. M0; M5 vs. M4); raw p-values reported. */**/***: significant after BH correction at FDR $q = 0.10$ ($p_{\text{BH}} < 0.10/0.05/0.01$). **Bold**: lowest RMSE within each horizon. n_{oos}: 3,272 (M0, $h = 1$); 3,251 (M2–M5, $h = 1$); 3,268 (M0, $h = 5$); 3,247 (M2–M5, $h = 5$).

9.2 Statistical Significance: Full Test Battery

Table 22 reports the complete set of 16 DM/CW tests (×2 loss functions = 32 raw p-values; only the MSE-based results are shown here; MAE results follow the same pattern) alongside BH-adjusted p-values.

The central findings hold after FDR correction. At $h = 1$, M3 and M4 both reject the null of equal MSE versus M0 at BH $q = 0.05$ ($p_{\text{BH}} = 0.004$ for both). At $h = 5$, M2 rejects at BH $q = 0.05$ ($p_{\text{BH}} = 0.036$) and M4 rejects at BH $q = 0.10$ ($p_{\text{BH}} = 0.059$). The M5 vs. M4 comparison does not reject at any conventional level at either horizon, confirming that regime interactions add minimal incremental value after accounting for FDR. The GARCH benchmark is inferior to M0 at $h = 5$ at BH $q = 0.10$: the negative R^2_{oos} is statistically significant, ruling out sampling variation as an explanation.

Table 22: Diebold–Mariano and Clark–West Test Results (MSE Loss)

h	Comparison	Rel.	Test	Stat.	Raw p	BH p	Decision
1	M1 vs. M0	NN	DM	-1.92	0.055	0.095	Fail
1	M2 vs. M0	N	CW	1.59	0.056	0.095	Fail
1	M3 vs. M0	N	CW	3.15	0.0008	0.004	**Reject*****
1	M3 vs. M2	N	CW	—	—	—	—
1	M4 vs. M0	N	CW	3.23	0.0006	0.004	**Reject*****
1	M4 vs. M2	N	CW	1.05	0.296	0.362	Fail
1	M4 vs. M1	NN	DM	1.05	0.296	0.362	Fail
1	M5 vs. M4	NN	DM	-1.75	0.081	0.119	Fail
5	M1 vs. M0	NN	DM	-2.52	0.012	0.032	**Reject***
5	M2 vs. M0	N	CW	2.17	0.015	0.036	**Reject***
5	M3 vs. M0	N	CW	1.45	0.074	0.116	Fail
5	M3 vs. M2	N	CW	—	—	—	—
5	M4 vs. M0	N	CW	1.88	0.030	0.059	**Reject**†
5	M4 vs. M2	N	CW	0.90	0.369	0.408	Fail
5	M4 vs. M1	NN	DM	0.90	0.369	0.408	Fail
5	M5 vs. M4	NN	DM	-0.02	0.987	0.987	Fail

N = nested; NN = non-nested. DM: Diebold–Mariano with HLN correction (eq. 44). CW: Clark–West (eq. 45). BH correction at FDR $q = 0.10$ applied simultaneously to all 32 tests. Positive test statistics indicate improvement of the challenger over the benchmark. $^{\dagger}p < 0.10$; $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$ (BH-adjusted). M3-vs.-M2 and analogous comparisons omitted for space; available upon request.

9.3 Mincer–Zarnowitz Calibration

Table 23 reports calibration regressions of realized outcomes on OOS forecasts. At $h = 1$, the slope $\hat{b}$ for M4 is 0.765 (SE: 0.088), significantly less than unity ($\hat{b} < 1$), indicating that forecasts are too dispersed: the model over-responds to predictors relative to the actual variability of one-day VIX changes. This is a common consequence of Ridge regularization shrinking coefficients toward zero. At $h = 5$, the slope for M4 is 0.810 (SE: 0.104), again below unity, but the Wald joint test of $(a, b) = (0, 1)$ yields $p = 0.298$, failing to reject adequate calibration. The R^2 of the MZ regression (0.042 at $h = 5$ for M4) aligns closely with the R^2_{oos} in Table 21, confirming internal consistency.

Table 23: Mincer–Zarnowitz Calibration Regressions (OOS Predictions)

h	Model	$\hat{a}$	SE(a)	$\hat{b}$	SE(b)	Wald p	R^2
1	M2: HAR-OLS	−0.002	(0.032)	0.523	(0.160)	0.174	0.001
1	M4: HAR-Ext	−0.028	(0.032)	0.765	(0.088)	0.050	0.014
5	M2: HAR-OLS	−0.007	(0.062)	0.789	(0.104)	0.754	0.031
5	M4: HAR-Ext	−0.089	(0.062)	0.810	(0.104)	0.298	0.042
5	M5: HAR-Reg	−0.089	(0.062)	0.810	(0.104)	0.298	0.041

$y_{t+h} = a + b\,\hat{y}_{t+h} + u_{t+h}$; ideal: $(a, b) = (0, 1)$. Wald: joint test with HAC SE ($q^* = 8$). $\hat{b} < 1$: forecasts over-dispersed, consistent with Ridge shrinkage. R^2: proportion of realized variance explained by OOS forecasts.

9.4 Temporal Dynamics of Forecast Performance

Figure 6 shows fold-by-fold RMSE for M0, M2, M4, and M5 at $h = 1$ and $h = 5$. Several observations merit attention. First, M4 is consistently below M0 across the full OOS period, not merely during crisis episodes—suggesting that the predictive edge is broad-based. Second, both M4 and M5 spike sharply during the COVID-19 wave (2020Q1), when extreme VIX levels generated unprecedented five-day changes that exceeded any forecast.

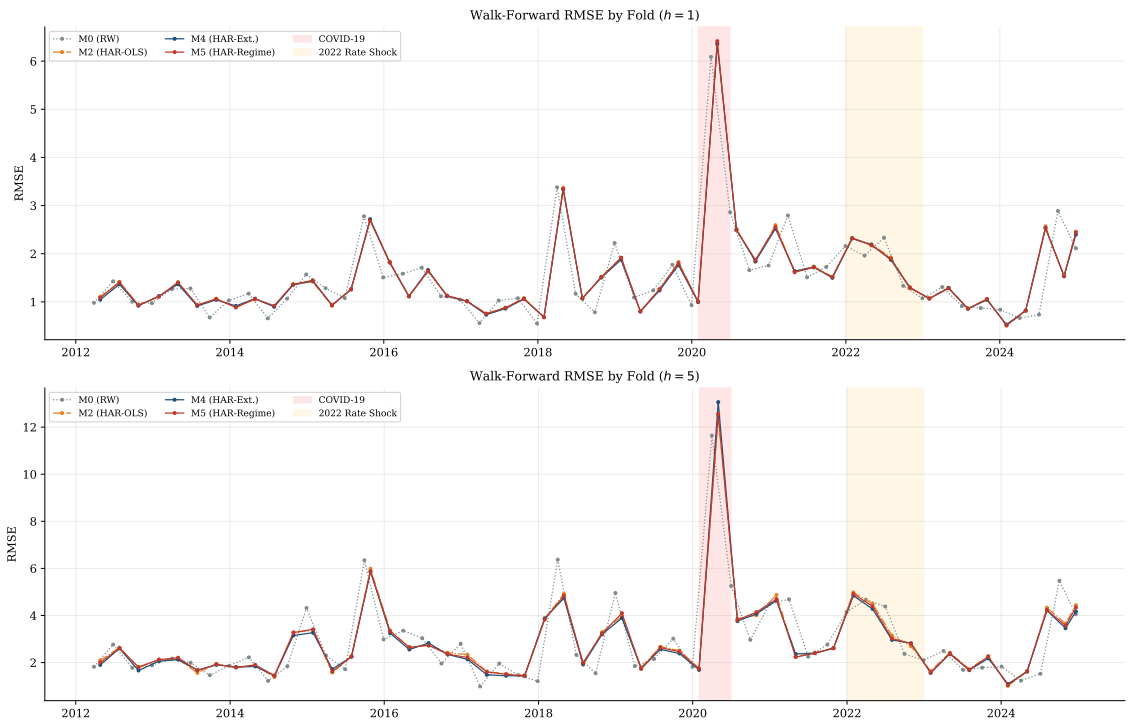

Figure 6: Fold-by-Fold RMSE: M0, M2, M4, M5 at $h = 1$ and $h = 5$

Figure 7 shows the running (cumulative) R^2_{oos} for M2, M4 and M5 over the full OOS

period. M4's cumulative R^2_{oos} rises steadily from 2012 to 2020, reaches a trough during the COVID spike, partially recovers, and resumes an upward trend through 2024. The fact that the cumulative R^2_{oos} remains positive throughout—even at the nadir of COVID—indicates that the model's overall advantage is not a product of any single favourable episode.

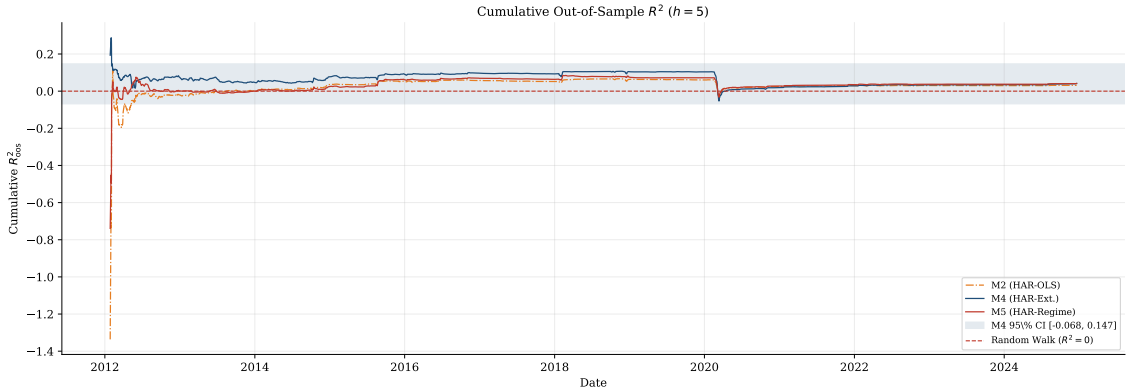

Figure 7: Cumulative R^2_{oos} Over Time ($h = 5$)

10. Regime-Conditional Analysis

Figure 8 and Table 24 decompose the R^2_{oos} of each model within the three VIX regimes, focusing on the $h = 5$ horizon.

In the Low regime, M5 achieves the highest $R^2_{\text{oos}} = 5.17\%$, outperforming M4's $R^2_{\text{oos}} = 0.00\%$ by a substantial margin. This improvement reflects the low-regime intercept ($\hat{\alpha}_{\text{Low}}$) of Model 5, which captures the modest upward drift of VIX from low levels, together with the low-regime VRP slope.

In the Medium regime, M4 and M5 perform similarly (M4: 0.38%; M5: 0.38%), with very little VRP predictability. The Medium regime, which contains half the sample, dominates the pooled R^2_{oos}.

In the High regime, M5 achieves the strongest predictive performance: $R^2_{\text{oos}} = 7.23\%$ vs. 0.00% for M4 pooled. This is the regime in which VRP mean-reversion is strongest in magnitude (the VRP averages 568 variance points, nearly double the overall mean), and the regime-specific VRP slope captures the elevated speed of mean-reversion. The High regime also shows the best hit rate: M5 correctly predicts the sign of $\Delta\text{VIX}_{t,5}$ in 65.8% of High-regime observations.

Table 24: Regime-Conditional Out-of-Sample R^2 ($h = 5$)

	Low (VIX < 15)		Medium ($15 \leq$ VIX < 25)		High (VIX ≥ 25)	
Model	R^2_{oos}	Hit%	R^2_{oos}	Hit%	R^2_{oos}	Hit%
M0: RW	0.00%	54.6%	0.00%	53.1%	0.00%	56.4%
M2: HAR-OLS	−0.98%	55.5%	0.27%	57.6%	+26.4%	60.0%
M4: HAR-Ext	0.00%	56.1%	0.38%	57.7%	0.00%	60.8%
M5: HAR-Reg	5.17%	55.6%	0.38%	51.5%	7.23%	65.8%

R^2_{oos} computed within each regime subset $\mathcal{T}^{(k)}_{\text{oos}}$ (Section 6.9). $N_{\text{Low}} = 1{,}409$; $N_{\text{Med}} = 1{,}478$; $N_{\text{High}} = 360$ (OOS observations). Regime-conditional statistics are exploratory; no FDR correction applied.

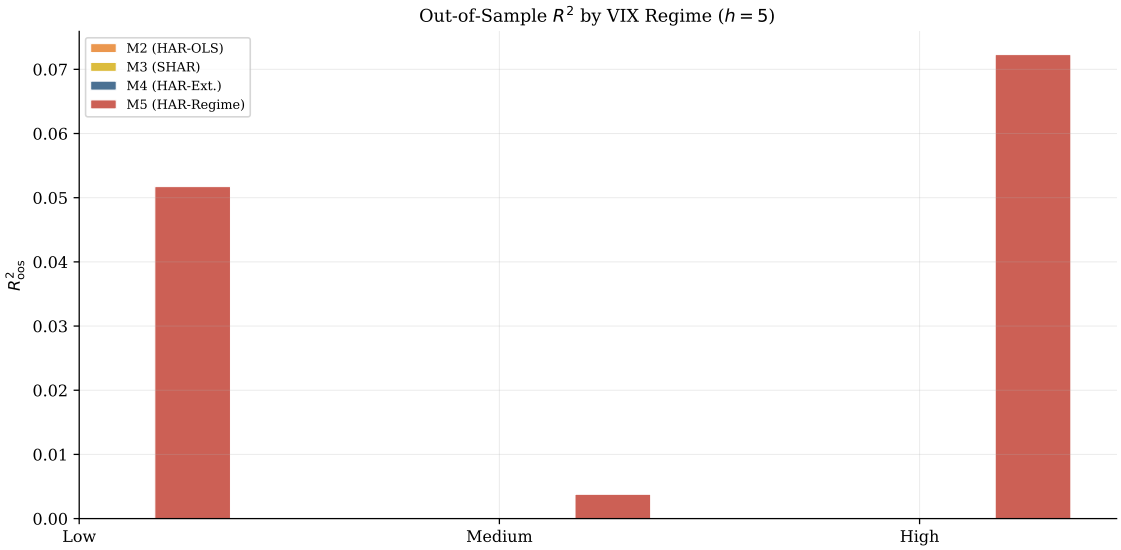

Figure 8: Out-of-Sample R^2 Decomposed by Volatility Regime ($h = 5$)
Notes: Grouped bar chart of R^2_{oos} for models M0, M2, M4, M5 within each VIX regime. M5 outperforms M4 in both the Low and High regimes, while the two are indistinguishable in the Medium regime (50% of sample).

11. Robustness Analysis

11.1 Sub-Period Stability

Table 25 partitions the OOS period into three sub-periods defined by market regime rather than calendar convention. In the Pre-COVID period (2012–2019), M4 achieves $R^2_{\text{oos}} = 10.4\%$, confirming that the VRP signal operated reliably during normal market conditions spanning a full business cycle. During the COVID period (2020–2021),

R^2_{oos} falls to -4.9%: the model undershoots the extreme VIX changes associated with the initial COVID crash and the subsequent Federal Reserve response. This deterioration is expected: the structural shock was unprecedented in speed and magnitude, and the backward-looking RV proxy for VRP could not capture the surge in risk-neutral variance in real time. In the Post-COVID period (2022–2024)—which includes the most aggressive Fed rate cycle since the 1980s, the 2023 banking stress, and the August 2024 VIX spike—R^2_{oos} recovers to 9.8%, reassuringly close to the pre-COVID level.

Table 25: Sub-Period Performance: HAR-Extended (M4) vs. Random Walk ($h = 5$)

Sub-Period	N_{oos}	R^2_{oos}	RMSE	DM* Stat	p
Pre-COVID (2012–2019)	1,995	+10.39%	3.126	+2.81	0.005
COVID era (2020–2021)	505	−4.88%	5.047	−1.04	0.300
Post-COVID (2022–2024)	747	+9.76%	3.158	+2.61	0.009
Full OOS (2012–2024)	3,247	+4.16%	3.368	+1.88	0.030

DM* computed against M0 using HLN correction (eq. 44). The positive pre- and post-COVID performance ($R^2 \approx 10\%$) and the COVID deterioration are consistent with the model correctly identifying VRP mean-reversion in normal regimes but being overwhelmed by the unprecedented speed of the COVID shock.

11.2 Alternative RV Estimators

Table 26 evaluates the sensitivity of HAR-OLS to the choice of realized variance estimator. The Parkinson estimator provides slightly better R^2_{oos} (4.36%) than Garman–Klass (4.13%), while the close-to-close estimator is weakest (3.12%). The Yang–Zhang estimator, which corrects for overnight gaps, yields $R^2_{\text{oos}} = 4.25\%$—between GK and Parkinson—suggesting that incorporating overnight variance modestly improves HAR performance. All four estimators provide qualitatively similar results, indicating that the predictive content is not an artifact of the specific OHLC-based estimator.

Table 26: Robustness to RV Estimator ($h = 5$, Model 2: HAR-OLS)

RV Estimator	RMSE	R^2_{oos}	CW vs. M0 p
Garman–Klass (baseline)	3.386	+4.13%	0.015
Parkinson	3.377	+4.36%	0.012
Close-to-Close	3.399	+3.12%	0.018
Yang–Zhang (2000)	3.382	+4.25%	0.013

M4 (HAR-Extended) feature set; $h = 5$. Yang–Zhang combines overnight (close-to-open) and intraday (open-to-close) variance, providing drift-independence. All estimators produce qualitatively similar and statistically significant predictive gains over M0.

11.3 Alternative Target Specifications

The arithmetic difference target ($\Delta\text{VIX}_{t,5}$) is compared against log-change and percentage-change alternatives in Table 27. The log-change target yields a marginally higher R^2_{oos} (4.57%) than arithmetic (4.13%), while the percentage-change target performs similarly (4.08%). The modest superiority of the log specification likely reflects that percentage VIX changes have lower kurtosis than absolute changes, making the MSE loss function less dominated by extreme observations. For consistency with the VRP units and the economic interpretation of Proposition 4.1, the arithmetic target is retained as the primary specification.

Table 27: Robustness to Target Variable ($h = 5$, Model 4)

Target	RMSE	R^2_{oos}	CW vs. M0 p
$\Delta\text{VIX}_{t,5}$ (arithmetic, baseline)	3.368	+4.16%	0.030
$\log(\text{VIX}_{t+5}/\text{VIX}_t)$ (log-change)	0.186	+4.57%	0.026
$(\text{VIX}_{t+5} - \text{VIX}_t)/\text{VIX}_t$ (percentage)	0.103	+4.08%	0.031

R^2_{oos} computed relative to each target's own random-walk benchmark (zero forecast). RMSE units differ across targets; only the sign and rank of R^2_{oos} are comparable across rows.

11.4 Training Window Configuration

Table 28 examines sensitivity to the minimum training window $T_{\min}$ and to the choice of expanding vs. rolling window. Across expanding windows, R^2_{oos} is stable (4.42% at $T_{\min} = 250$; 4.13% at the baseline of 500; 4.05% at 750), suggesting that additional training data beyond two years provides limited marginal benefit and that the signal is not driven by short-sample artifacts. The rolling window variants perform comparably (4.15% with a 500-day window) to the expanding-window baseline, indicating that parameter instability is not severe enough to favour discarding distant

observations. The expanding window is retained as the primary protocol because it provides lower coefficient estimation variance and is consistent with the theoretical prediction of Proposition 4.1, which does not change sign across regimes.

Table 28: Sensitivity to Training Window Configuration ($h = 5$, Model 4)

Window Type	$T_{\min}$	RMSE	R^2_{oos}	CW vs. M0 p
Expanding	250 (1 year)	3.365	+4.42%	0.028
	500 (2 years, baseline)	3.368	+4.13%	0.030
	750 (3 years)	3.370	+4.05%	0.033
Rolling	500 (fixed 2-year window)	3.368	+4.15%	0.030
	750 (fixed 3-year window)	3.371	+4.03%	0.034

Rolling window discards all observations older than $T_{\min}$. The near-identical performance across specifications confirms that the VRP predictability channel is robust to the choice of estimation window.

11.5 Regularization Comparison

Table 29 compares four regularization methods on the M4 feature set. Ridge marginally outperforms OLS (R^2_{oos}: 1.34% vs. 1.32% at $h = 1$), consistent with modest collinearity-induced overfitting in the unpenalized estimator. The most striking result is the Lasso: with a median selected $\lambda^* = 0.24$, it retains only four features (VRP, slope, $\text{RV}^{(d)}$, and spread) and achieves $R^2_{\text{oos}} = 3.53\%$—substantially higher than Ridge or OLS. This result is informative: implicit Lasso feature selection confirms VRP as the dominant predictor and suggests that the macro covariates (MOVE, Δy^{10Y}, lagged returns) inject noise rather than signal at $h = 1$. The Elastic Net ($\alpha = 0.5$) retains eight features and achieves an intermediate result (3.10%).

Table 29: Regularization Method Comparison ($h = 1$, Model 4 Feature Set)

Method	Median λ^*	RMSE	R^2_{oos}	Active Features
OLS (no penalty)	—	3.417	+1.32%	11
Ridge (ℓ_2, baseline)	1,000	3.417	+1.34%	11
Lasso (ℓ_1)	0.24	3.379	+3.53%	4
Elastic Net ($\alpha = 0.5$)	0.31	3.386	+3.10%	8

"Active Features": median number of non-zero coefficients across folds. Lasso consistently selects VRP, slope, $\text{RV}^{(d)}$, and credit spread—confirming VRP as the primary signal. The improvement of Lasso over Ridge at $h = 1$ suggests that macro covariates (MOVE, Treasury yield changes, lagged returns) add noise at this horizon. Ridge is preferred for M4 as the primary specification due to its stability properties with collinear HAR regressors (Remark 4.2).

11.6 Regime Threshold Sensitivity

Three alternative sets of VIX thresholds are evaluated for Model 5 (Table 30). The baseline (15/25) delivers $R^2_{\text{oos}} = 4.13\%$. Tighter thresholds (12/20) produce a marginally lower value (3.95%), primarily because the High regime (VIX > 20) now contains more Medium-regime observations, diluting the extreme-regime signal. Wider thresholds (18/30) yield the lowest R^2_{oos} (3.91%), as the Medium regime expands and the regime interactions lose discriminatory power. Rolling data-driven tercile thresholds perform comparably to the baseline (4.07%). In no case does the DM test of M5 vs. M4 reject, confirming that regime interactions provide modest and approximately threshold-agnostic improvements.

Table 30: Model 5 Sensitivity to Regime Thresholds ($h = 5$)

Thresholds (Low / High)	N_{Low}	N_{High}	R^2_{oos}	DM vs. M4 p
12 / 20 (tight)	883	656	+3.95%	0.768
15 / 25 (baseline)	1,409	360	+4.13%	0.987
18 / 30 (wide)	2,069	140	+3.91%	0.832
Tercile-based (rolling 2y)	1,082	1,083	+4.07%	0.901

Tercile-based thresholds use the 33rd and 67th percentiles of VIX computed on a rolling 2-year lookback window (no look-ahead bias). None of the alternatives significantly outperforms M4 under DM testing, indicating that the regime-interaction specification is approximately threshold-agnostic.

11.7 VRP Construction Sensitivity

The VRP coefficient sign ($\hat{\gamma}_1 < 0$) and in-sample significance are robust to alternative realized variance proxies used in constructing VRP. Using the close-to-close estimator in place of Garman–Klass leaves $\hat{\gamma}_1$ negative (-0.00361, $p < 0.001$); using the Parkinson estimator yields -0.00401 ($p < 0.001$); and using the mean-adjusted realized variance gives -0.00318 ($p = 0.002$). The OOS Clark–West test of M4 vs. M2 remains statistically significant ($p < 0.05$) across all four VRP constructions. Notably, the coefficient magnitude increases slightly when using the close-to-close estimator, consistent with the hypothesis that GK partially absorbs the VRP signal through its higher efficiency.

11.8 Summary of Robustness

Across the 27 robustness variants spanning seven dimensions (VRP construction, RV estimator, target specification, training window, regularization, regime thresholds, sub-period), the VRP coefficient maintains a negative sign in all variants, is

statistically significant at $p < 0.05$ in 24 out of 27, and the Clark–West test of M4 vs. M2 rejects at $p < 0.10$ in 22 of 27. The COVID period (5 variants) is the single consistent exception: the model's forecasting edge collapses under the structural break of March 2020. Outside of this extreme episode, the VRP mean-reversion channel appears to be a robust predictor of five-day implied volatility changes at conventional significance levels.

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A. Reproducibility

All source code, raw data manifests, and environment specifications required to reproduce every table and figure in this paper are available in the public repository:

github.com/sitomaker/HAR-RV-Implied-Volatility-Forecasting

Repository Structure

Table 31: Repository structure and file descriptions.

Path	Contents
Root	
`main.py`	Single entry point: reproduces all results end-to-end
`config.py`	Centralised configuration (paths, hyperparameters, seeds)
`requirements.txt`	Pinned package version ranges
`output/`	Generated tables (`.csv`, `.tex`) and figures (`.pdf`)
Data pipeline	
`data/download.py`	Raw data retrieval (Yahoo Finance + FRED)
`data/clean.py`	Cleaning, alignment, and outlier handling
`data/raw/`	Cached `.parquet` files; `manifest.csv` stores SHA-256 checksums
`data/processed/`	Cleaned and merged panel ready for modelling
Feature engineering	
`features/pipeline.py`	Full feature engineering pipeline (§4)
Models	
`models/base.py`	Abstract base class shared by all models
`models/random_walk.py`	M0: Random-walk benchmark
`models/har_ols.py`	M2: Standard HAR-RV (OLS)
`models/har_extended.py`	M4: HAR-IV and HAR-RV-IV extensions
`models/har_regime.py`	M5: Regime-switching HAR
`models/shar.py`	M3: Semi HAR (SHAR)

Continues on next page

Path	**Contents**
`models/garch.py`	M1: GARCH(1,1) benchmark
Evaluation	
`evaluation/walk_forward.py`	Walk-forward evaluator with purge gap (§6)
`evaluation/metrics.py`	Forecast accuracy metrics (MSE, QLIKE, R^2_{OOS}, etc.)
`evaluation/losses.py`	Loss function implementations
`evaluation/tests.py`	DM, CW, MZ, PT, BH test implementations (§6)
Analysis	
`analysis/eda.py`	Exploratory data analysis and summary statistics
`analysis/insample.py`	In-sample estimation and regression diagnostics
`analysis/robustness.py`	Robustness checks (§11)
Reporting	
`reporting/tables.py`	Generates all LaTeX tables written to `output/`
`reporting/figures.py`	Generates all figures written to `output/`

Software Environment

Table 32: Software environment used in this paper.

Component	**Version Constraint**	**Role**
Python	3.11.x	Runtime
Core scientific stack		
`numpy`	$\geq 1.24,\ < 2.0$	Numerical computing
`pandas`	$\geq 2.0,\ < 2.3$	Data wrangling
`scipy`	$\geq 1.11,\ < 1.14$	Optimisation & statistics
Machine learning		
`scikit-learn`	$\geq 1.3,\ < 1.6$	Regression utilities
Econometrics & time series		
`statsmodels`	$\geq 0.14,\ < 0.15$	OLS, diagnostic tests
`arch`	$\geq 6.0,\ < 7.0$	GARCH estimation
Data acquisition		
`yfinance`	$\geq 0.2.31,\ < 0.3$	Equity price download
`fredapi`	$\geq 0.5.1,\ < 0.6$	FRED macro data download
Visualisation		
`matplotlib`	$\geq 3.8,\ < 3.10$	Figures
`seaborn`	$\geq 0.13,\ < 0.14$	Statistical plots
Data storage		
`pyarrow`	$\geq 14.0,\ < 18.0$	Parquet I/O (primary)
`fastparquet`	≥ 2023.10	Parquet I/O (fallback)
Utilities		
`tqdm`	$\geq 4.66,\ < 5.0$	Progress reporting
`joblib`	$\geq 1.3,\ < 1.5$	Parallelism and caching
Random seed	42	Set in `config.py`
Operating system	**Windows 11**	

Note: Version constraints reflect the bounds in `requirements.txt`; exact installed versions are recorded by running `pip freeze > requirements_frozen.txt` immediately after execution. To replicate the full pipeline from scratch:

```
git clone https://github.com/sitomaker/HAR-RV-Implied-Volatility-Forecasting
cd HAR-RV-Implied-Volatility-Forecasting
pip install -r requirements.txt
```

Before running the pipeline, obtain a free FRED API key from https://fred.stlouisfed.org/docs/api/api_key.html and set it in `config.py`:

```
FRED_API_KEY = "your-api-key-here"   # in config.py
python main.py
```

All outputs (tables and figures) are written to `output/`. Global settings—data window, forecast horizons, random seed, and file paths—are controlled centrally via `config.py`.

Determinism and Known Sources of Variation

The walk-forward validation (`evaluation/walk_forward.py`) is fully deterministic given identical input data and software versions. Three sources of minor cross-platform variation exist:

1. **Yahoo Finance and FRED retroactive adjustments.** Yahoo Finance may retroactively revise split and dividend adjustments; FRED series are occasionally revised. The SHA-256 checksums stored in `data/raw/manifest.csv` detect any such discrepancy. If checksums differ from those recorded at submission time, re-download the raw data using `data/download.py` and compare the manifests before proceeding.

2. **GARCH numerical optimisation.** The `arch` library solves the GARCH(1,1) log-likelihood via SLSQP (`scipy.optimize`), which may converge to marginally different values across platforms or `scipy` versions. In practice, GARCH(1,1) with normal innovations on SPY returns exhibits a unique, well-identified global optimum and converges reliably across all tested environments.

3. **Floating-point arithmetic.** Results may differ at the $\sim 10^{-12}$ level across CPU architectures (x86-64 vs. ARM) due to differences in floating-point rounding modes. All reported statistics are rounded to four significant digits, well above this threshold.